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EXOGENOUS OIL SUPPLY SHOCKS: HOW BIG ARE THEY AND HOW MUCH DO THEY MATTER FOR THE U.S. ECONOMY?

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Abstract—The paper proposes a new measure of exogenous oil supply shocks. The timing, the magnitude, and the sign of this measure may differ greatly from current state-of-the-art estimates. It is shown that only a small fraction of the observed oil price increases during oil crisis periods can be attributed to exogenous oil production disruptions. Exogenous oil supply shocks cause a sharp drop of U.S. real GDP growth after five quarters rather than an immediate and sustained reduction in economic growth and a spike in CPI inflation after three quarters. Overall, exogenous oil supply shocks made remarkably little difference for the evolution of the U.S. economy since the 1970s, although they did matter for some historical episodes.

I. Introduction

SINCE the oil crises of the 1970s there has been strong interest in the question of how oil production shortfalls caused by wars and other exogenous political events in OPEC countries affect oil prices, U.S. real GDP growth, and U.S. CPI inflation (see, for example, Barsky & Kilian, 2002, 2004; Bernanke, Gertler, & Watson, 1997; Hamilton, 1983, 1996, 2003; Hoover & Perez, 1994). This paper focuses on three key questions: How large are the exogenous fluctuations in the production of oil? What are the dynamic effects of exogenous oil production shortfalls on U.S. real GDP growth and CPI inflation? To what extent do exogenous oil supply shocks explain changes in the real price of oil?

I introduce a new methodology for quantifying the shortfall of OPEC oil production caused by exogenous political events such as wars or civil disturbances. This methodology utilizes monthly production data for all OPEC countries and data on aggregate non-OPEC oil production that are available from the U.S. Department of Energy since January 1973. It is based on the observation that any attempt to identify the timing and magnitude of these exogenous production shortfalls requires explicit assumptions about the counterfactual path of oil production in the absence of the exogenous event. The strategy is to generate the counterfactual production level for the country in question by extrapolating its prewar production level based on the average growth rate of production in other countries that are subject to the same global macroeconomic conditions and economic incentives, but are not involved in the war. Which countries belong in this benchmark group must be decided on a case-by-case basis drawing on historical accounts and industry sources. This approach allows the construction of a monthly time series of exogenous oil production shortfalls since 1973. The change over time in this exogenous production shortfall series (expressed as a percentage share of world oil production) provides a natural measure of the exogenous oil supply shock.

Measures of exogenous oil production shortfalls allow the estimation of the dynamic effects of exogenous fluctuations in oil production on macroeconomic aggregates such as real GDP growth. One possibility is to use measures of the exogenous oil production shortfall as instruments in regressions that project macro aggregates on the price of oil, as suggested by Hamilton (2003). As shown in Kilian (2005), such instrumental variable (IV) estimates tend to suffer from a weak-instrument problem, calling into question the reliability of the empirical results. I propose an alternative, simpler approach to estimating the dynamic effects of exogenous oil supply shocks that does not require IV estimation, but only involves ordinary least squares (OLS) projections and standard methods of inference. This alternative method involves projecting the macroeconomic variable of interest on a constant, lagged values of the dependent variable, and the current and lagged values of the exogenous oil supply shock. It provides a direct estimate of the dynamic effects of exogenous oil supply shocks on macroeconomic aggregates.

Using this new approach, I find statistically significant evidence of a sharp drop in real GDP growth five quarters after an exogenous oil supply shock and of a spike in CPI inflation three quarters after the shock. I also study the question of how these macroeconomic aggregates would have evolved in the absence of exogenous oil production shocks. I show that the effects of exogenous oil supply shocks on U.S. real GDP growth and CPI inflation were comparatively small on average, but that they did matter for particular historical episodes such as the 1990/91 Persian Gulf War. These empirical conclusions are robust to many changes in the counterfactual underlying the oil supply shock measure.

In addition, I investigate the predictive content of exogenous oil supply shocks for changes in the real price of oil. My analysis suggests that exogenous oil production shortfalls are of limited importance in explaining oil price changes during crisis periods. This result is robust to the choice of the exogenous oil supply shock measure. Of the episodes studied, only the 1980/81 oil price increases can be attributed to exogenous oil supply disruptions. Exogenous oil supply disruptions were an important contributing factor for the oil price increases in the third quarter of 1979, but do not help explain the subsequent oil price increases during

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that episode. Exogenous oil supply disruptions explain only a small fraction of the oil price increases during the 1973/74, 1990/91, and 2002/03 episodes. This finding is suggestive of an important role for other explanatory variables such as shifts in the demand for oil or shifts in the uncertainty about future oil supply shortfalls that are unrelated to actual production shortfalls. The latter explanation is shown to explain the bulk of the oil price fluctuations during the 1990/91 Persian Gulf War episode. I also provide evidence that the surge in global economic growth in 1973, 1979, and 2003/04 coincided with capacity constraints in oil production, providing a plausible alternative explanation of the rapid oil price increases observed during these episodes.

The results in this paper differ from the conventional wisdom in the literature along a number of dimensions. First, I find that the timing, magnitude, and even the sign of exogenous oil supply shocks may differ greatly from current state-of-the-art estimates. Second, the results in this paper differ markedly from the common view that "the case for exogeneity of at least the major oil price shocks is strong" (Bernanke, Gertler, & Watson, 1997, p. 93). The evidence suggests that this view is supported by the data for the 1980/81 and 1990/91 oil price shocks, but not for other oil price shocks. I also show that statistical measures of the net oil price increase relative to the recent past do not represent the exogenous component of oil prices. In fact, only a small fraction of the observed oil price increases during crisis periods can be attributed to exogenous oil production disruptions. Third, compared with previous indirect estimates of the effects of exogenous supply disruptions on real GDP growth that treated major oil price increases as exogenous, my direct estimates suggest a sharp but temporary reduction in real growth after about one year rather than an immediate and sustained reduction in economic growth. I also find a one-time spike in CPI inflation in response to an exogenous oil supply contraction rather than a sustained increase in inflation, as is sometimes conjectured. Finally, my results put into perspective the importance of exogenous oil production shortfalls in the Middle East. A counterfactual historical analysis suggests that exogenous oil supply shocks made remarkably little difference overall for the evolution of U.S. real GDP growth and CPI inflation since the 1970s, although they did matter for some historical episodes. This result is driven by the fact that most exogenous oil supply shortfalls were at least partially reversed over time.

The remainder of the paper is organized as follows. Section II briefly reviews the exogenous oil supply shock dates since 1973. Section III proposes explicit counterfactuals for each of these events that allow the construction of monthly and quarterly time series of the exogenous OPEC oil production shocks. Section IV uses these data to assess the impact of exogenous variation in the supply of oil on U.S. CPI inflation and real GDP growth. This section presents dynamic responses to exogenous oil supply shocks as well as counterfactual simulations of the path of U.S. macroeconomic aggregates following exogenous oil supply shocks. Section V contrasts the approach to constructing explicit measures of exogenous oil supply shocks introduced in this paper with the commonly used quantitative dummy variable approach (see Hamilton, 2003). I show that the choice of oil supply shock measure matters for the qualitative patterns of the responses of real GDP growth, real GDP levels, and CPI inflation. Section VI explores the tenuous link between exogenous oil supply shocks and oil price shocks. I show that standard measures of oil price shocks based on nonlinear transformations of oil price data do not identify the exogenous component of oil price changes, as is sometimes claimed. I also compare the explanatory power of alternative measures of exogenous oil supply shocks for the change in the real price of oil and show that none of these measures does a good job at predicting changes in real oil prices. Given this evidence, I examine alternative explanations of oil price shocks based on precautionary demand shifts and based on shifts in the global demand for industrial commodities (including oil). In this context, I highlight the important role played by capacity constraints in crude oil production. Section VII concludes.

II. A Review of Exogenous Oil Shock Dates since 1973

The analysis of oil supply shocks in this paper focuses on the modern OPEC period that began in late 1973. The sample ends in September 2004. The objective is to identify and quantify shocks to OPEC oil production during this period that are exogenous with respect to U.S. macroeconomic aggregates. Natural candidates for such shocks are the oil production disruptions caused by the Iranian Revolution of 1978/79, by the Iran-Iraq War of 1980–1988, by the Gulf War of 1990/91, and by the Iraq War of 2003. A good case can be made that the production cutback caused by the civil unrest in Venezuela in 2002/03 represents another such shock, as shown in section III.[1] In contrast, there is no evidence that would suggest the occurrence of a major exogenous oil supply shock in Nigeria.[2]

Other studies of exogenous oil supply shocks after January 1973 have typically included in addition the Yom

[1] Based on Hamilton's (2003) methodology, the last two events were each associated with a 3.5% reduction in world oil supply. Although individually those shocks are only about half the size of the earlier shocks, they occurred within a few months of one another. By this metric, their joint effect should be of the same magnitude as that of the Iran-Iraq War, for example, and only slightly smaller than the Persian Gulf War, the Iranian Revolution, or the October 1973 war and subsequent oil embargo.

[2] Clearly, there are more exogenous political events in the region such as the Yemeni civil war from May to July 1994, the Egyptian-Libyan War of July 1977, or Israel's invasion of Lebanon in June 1982. What distinguishes the events we focus on from the latter examples is that they are thought to have caused a physical disruption of oil production in OPEC countries.

Kippur War of October 1973 (which was followed by the Arab oil embargo from October 1973 to March 1974). It is not clear whether this 1973/74 episode should be treated as an exogenous shock to oil production along with events such as the Iranian Revolution or the Iran-Iraq War. As I will discuss below, no OPEC oil facilities were attacked during the October 1973 war, and there is no evidence of OPEC production shortfalls caused by military action. While the subsequent "Arab oil embargo" undoubtedly coincided with a major fall in Arab OPEC oil output (the extent of which I will quantify below), the question of whether this drop can be regarded as exogenous with respect to U.S. macroeconomic conditions remains a topic of debate (see Barsky & Kilian, 2002, and Hamilton, 2003, for further discussion). This is not the place to review the arguments for and against the exogeneity of the 1973/74 oil embargo. Instead I will present baseline results that include this event among the exogenous events, consistent with earlier studies. I also computed additional results that exclude this event. My qualitative results are not affected by whether the production cutbacks associated with the 1973/74 Arab oil embargo are treated as exogenous or not.

III. Constructing a Counterfactual

The proposed methodology utilizes monthly production data for all OPEC countries and for aggregate non-OPEC oil production that are available from the U.S. Department of Energy since January 1973. Consider an exogenous event such as a war (or civil disturbance) that lowers oil production in a given OPEC country. The proposal is to generate the counterfactual production level for this country by extrapolating its production level in the month right before the event in question based on the average growth rate of production in other countries that are subject to the same global macroeconomic conditions and economic incentives, but are not directly affected by that event. Which countries belong in this benchmark group must be decided on a case-by-case basis drawing on historical accounts and industry sources. The proposed method allows the construction of a monthly time series of exogenous oil production shortfalls that takes full account of the timing of the exogenous production shortfall and of variations over time in its magnitude. In sections IIIA–IIIF, I illustrate the implementation of this approach for the exogenous events listed in section II, starting with the 1973/74 event. In section IIIG, I construct a measure of the aggregate exogenous production shortfall across all OPEC countries. The change over time in this exogenous production shortfall series (expressed as a percentage share of world oil production) provides a natural measure of the exogenous oil supply shock.

A. *Counterfactual for the 1973 Arab-Israeli War and the 1973/74 Arab Oil Embargo*

The 1973/74 oil shock episode involves three conceptually distinct components: (i) any unplanned production shortfall caused by military action during the October 1973 Arab-Israeli War; (ii) the oil export embargo of Arab oil producers targeted against selected OECD countries that were viewed as pro-Israel; and (iii) the deliberate production cutbacks implemented by some Arab oil producers toward the end of 1973. Of these three distinct components only the last one matters for our purposes.

Although some of the decline in OPEC oil production after September 1973 occasionally is attributed to the destruction of oil facilities during that war, there is little evidence for that view. The weekly *Oil and Gas Journal* on October 15, 22, and 29 reported in detail about war-related damage to oil facilities in the region, notably in Syria, Lebanon, Israel, and Egypt. No damage in Iraq or any other OPEC country was reported. Thus, we can rule out the notion that, by destroying oil facilities in OPEC countries, the war caused an unplanned production shortfall.[3] Similarly, the politically motivated embargo against specific countries was not by itself the cause of a reduction in oil supplies. Although shipments of oil to some oil-importing countries were initially conditioned on specific diplomatic demands, it was quickly understood by all sides that such targeting would not and could not be effective because oil can be resold or simply diverted (see Terzian, 1985, pp. 178–180, for examples). For that reason I focus on the production cutbacks that took place during the Arab oil embargo between October 1973 and March 1974 as a result of deliberate policy decisions. In assessing the effect of this policy, the key question is how large this temporary production shortfall actually was. The construction of a counterfactual involves, first, an assessment of what normal production levels would have been prior to the oil shock. Second, it involves an assumption about how production would have developed in the absence of the war and embargo.

It is well known that oil production from Arab OPEC countries fell between September and November of 1973, whereas oil production in the rest of the world did not. It may seem natural at first to attribute the entire differential to the effects of political events in the Arab world. That approach would be misleading, however, because Middle Eastern OPEC oil producers were subject to different eco-

[3] Although there is no evidence of a reduction in OPEC oil output directly caused by the war, there was a reduction in Iraqi oil exports to the Mediterranean when the Tripoli oil terminal in Lebanon was attacked in October 1973. The *Oil and Gas Journal* conjectured that the prewar export volume on that route may have been as high as 0.500 million barrels per day. The reduction in Iraqi production in October is 0.320 million barrels per day relative to September. It is unclear to what extent that production cut was related to the Tripoli attack.

nomic incentives than producers in the rest of the world in the period leading up to October 1973. The difference in incentives arose as a result of the 1971 Tehran/Tripoli agreements between the oil companies and Middle Eastern OPEC oil producers. These five-year agreements in short provided a moderate improvement in the financial terms that host governments received from oil companies for each barrel of oil extracted by the oil companies in exchange for assurances that these governments would allow the oil companies to extract as much oil as they saw fit on those terms (see Seymour, 1980, p. 80).[4] The latter option principally affected Saudi Arabia as the country with the largest spare capacity. As reported by the *Oil and Gas Journal* of November 12, 1973, projections of future Saudi oil production under the leadership of U.S. oil companies up to October 1973 envisioned ever-expanding oil production "based on the premise that this is how much oil the Middle East would need to produce to balance the world's oil demand."

When global demand for oil accelerated in 1972/73, reflecting a worldwide economic boom (see Seymour, 1980, p. 100), many Middle Eastern countries were operating close to capacity already and were unable to increase oil output; whereas others, notably Saudi Arabia, had the capacity to increase oil output and did increase output prior to October 1973, albeit reluctantly. This reluctance can be attributed to the fact that the posted price agreed upon in 1971 might have been reasonable at the time, but was quickly eroded in real terms as a result of dollar devaluations and rising U.S. inflation. This development caused increasing opposition to the Tehran/Tripoli agreements that intensified in March of 1973 and culminated in the repudiation of the agreements by the oil-producing countries in early October of 1973. According to the *Oil and Gas Journal* article, the Saudi refusal to supply virtually limitless quantities of oil after September 1973, though perhaps prompted by the Mideast war and embargo, had "little to do with politics," but with the fact that Saudi Arabia in October 1973 had become independent of foreign oil companies and for the first time was able to make a "rational decision" about her production levels. In other words, with the repudiation of the Tehran/Tripoli agreements Saudi production levels should have reverted to normal levels, consistent with the level of production in non-OPEC countries, even in the absence of an embargo.

In this view, a substantial fraction of the observed decline in Arab oil output in late 1973, notably in Saudi Arabia and Kuwait, was simply a reversal of an unusual increase in Arab oil production (relative to non-OPEC oil producers) earlier that year in response to pressure from the oil companies. Moreover, the decision to reduce oil production was clearly motivated by the cumulative effects of the dollar devaluation, unanticipated U.S. inflation, and strong global economic growth, making it endogenous with respect to global macroeconomic conditions.

If this view is right, one would expect to see reductions in oil production in late 1973 only in those countries that expanded output in early 1973, but not in the other countries. Only to the extent that there are additional cuts beyond the production level that prevailed before the expansion, the case could be made that these cuts were politically motivated. This conjecture may be verified empirically. A natural benchmark against which to judge the validity of this explanation is the oil production of non-OPEC countries over the same time period. The upper panel of figure 1 shows the number of extra barrels produced per day in selected OPEC countries since March 1973 relative to the production level one would have expected, had those countries' production grown at the same rate as non-OPEC oil production.[5] Figure 1 confirms that Saudi Arabia and to a lesser extent Kuwait produced disproportionately more oil after March 1973 than other countries. Their output peaked in mid-1973. Thus, as conjectured earlier, all of the reduction in Saudi and Kuwaiti oil production in October 1973 (and some of the decline in November) can be understood as a return to normal production levels by international standards. Only the fall in oil production below the counterfactual in November and in December of 1973 is prima facie evidence of an exogenous oil production shortfall. In contrast, the other Arab members of OPEC combined never experienced an unusual increase in oil production in early 1973.[6] Thus, the decline in their oil output relative to international levels that started in November 1973 can be attributed in its entirely to political factors.

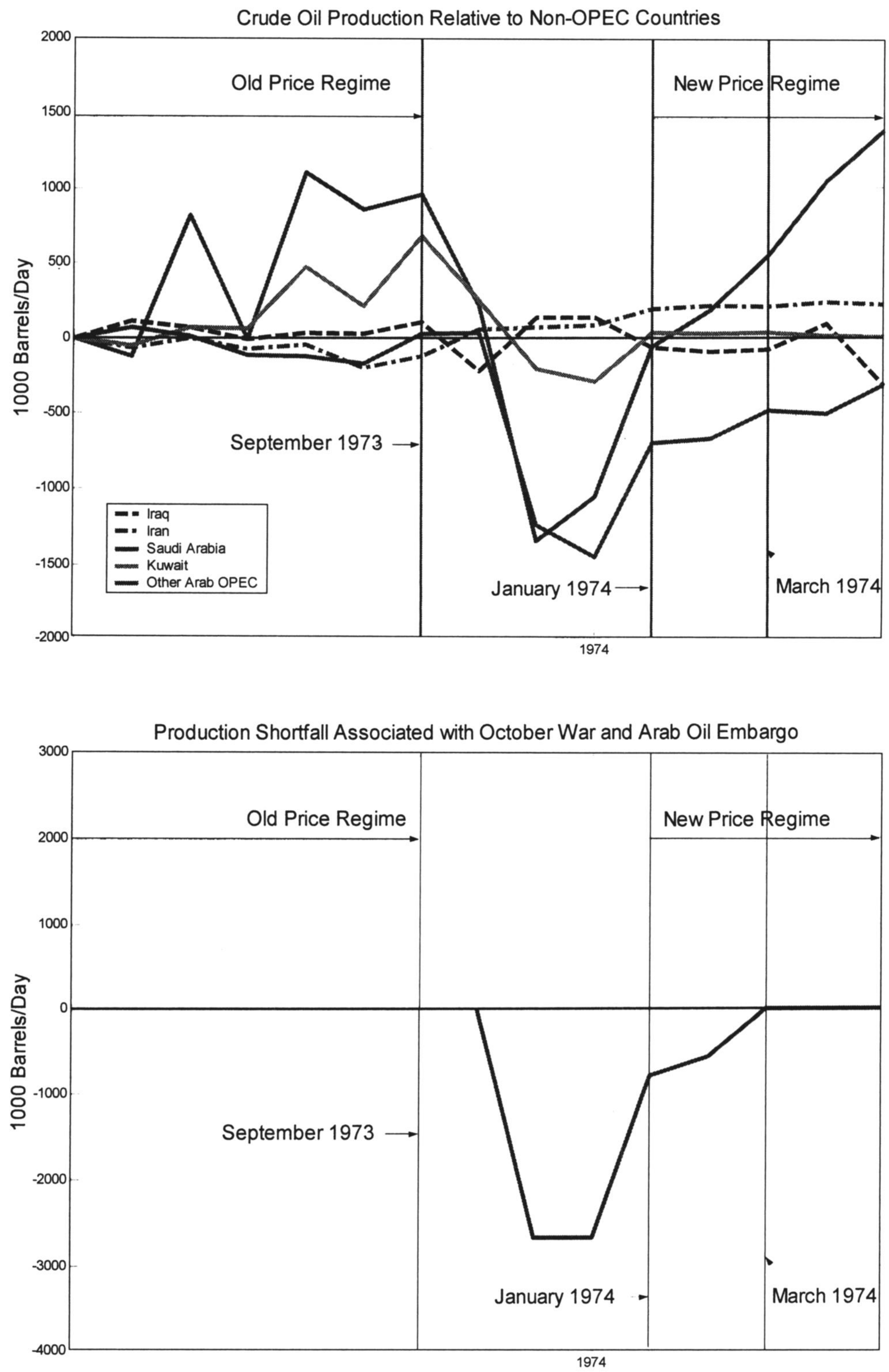
The data in figure 1 also show that the Saudi and Kuwaiti embargo was effectively over by January 1974 (when the price of oil stabilized at its new, much higher level). Among other Arab OPEC countries the cutback persisted much longer, yet at diminishing levels. In both cases, the bulk of the output reduction is concentrated in November and December. Finally, figure 1 shows that Iraq never was a major contributor to the production shortfall associated with the oil embargo, consistent with its public rhetoric (see Seymour, 1980, p. 119).

[4] Wilkins (1976, p. 168) cites the 1974 testimony of a representative of Exxon before the U.S. Senate that the oil companies in Tehran "made every effort to minimize posted price increase and to assure some security of supply."

[5] March 1973 is a natural starting date, since at this point the pressure to abandon the existing oil market regime intensified, suggesting that OPEC countries were forced to produce more than they would have wanted to at the prevailing price of oil. I also explored earlier starting dates; the results are qualitatively similar, but the production shortfall associated with the embargo is smaller than under the March scenario.

[6] This result also holds when considering each of these countries separately.

FIGURE 1.—THE OCTOBER 1973 WAR AND 1973/74 EMBARGO



Since the production levels in figure 1 are normalized relative to non-OPEC output, they already account for the increase in output stimulated by the initial increase in oil prices in early October. Since the increase in prices announced in early October 1973 had been determined prior to the war and was motivated on economic grounds, the case can be made that it would have been announced in any case (see, for example, Seymour, 1980, pp. 98, 107, and 113).[7] This interpretation is also consistent with the participation of Iran, a non-Arab country that was not a participant in the war, in the early October oil price increase and with the fact that the price increase was adopted by other non-Arab OPEC countries such as Venezuela and Nigeria.[8]

The lower panel of figure 1 shows the total production shortfall that is associated with political events in the Arab world in late 1973. The totals shown represent the contributions of all Arab OPEC countries, that is, Saudi Arabia, Kuwait, the United Arab Emirates, Qatar, Libya, Algeria, and Iraq. There is no evidence of a production shortfall in October 1973. At its peak in November and December the embargo involved an output reduction of about 2.67 million barrels per day. In January and February, the shortfall drops to 0.80 and 0.57 million barrels per day. Thereafter, the shortfall is 0.

B. *Counterfactual for Iran*

The beginning of the Iranian Revolution can be dated in October 1978, when striking workers paralyzed oil installations across the country. Although the abdication of the Shah and the transition to the Khomeini government was complete in January 1979, the effects of the revolution on Iranian oil production continue even today. How would Iranian production have developed in the absence of the Iranian Revolution? It is natural to restrict the benchmark to other OPEC countries operating under similar economic incentives. As discussed in sections IIIC and IIIE, Iraq and Saudi Arabia themselves experienced exogenous production disturbances as a result of the Iranian Revolution that necessitate their exclusion from the benchmark. Thus, the growth of OPEC production excluding Iran, Iraq, and Saudi Arabia provides a natural benchmark against which to compare the actual growth in Iranian oil production immediately after the revolution. This benchmark allows us to construct a counterfactual path for Iranian oil production from October 1978 until August 1980.

An obvious concern is that these benchmark countries might have increased production in response to higher oil prices triggered by the Iranian oil supply cuts. Without addressing the question of the extent to which prices rose as a result of this event, this concern may be dealt with by noting that, as in September 1973, these countries were already operating close to their capacity limit and thus were unable to increase production much further. While there are no data on capacity utilization rates by OPEC country, data from the International Energy Administration (IEA) confirm that utilization rates were at or near peak levels in 1979 both in OPEC and worldwide, when oil prices rose sharply. This view is further supported by the fact that actual oil production in the proposed benchmark countries hardly increased after the Iranian Revolution (and in fact started falling soon thereafter), as shown in figure 2.

The outbreak of the Iran-Iraq War in September 1980 does not require any modification in this benchmark since we already have excluded all countries whose production was directly affected by this second exogenous shock. The counterfactual for Iran continues to be the behavior of total OPEC production minus the combined production of Iran, Iraq, and Saudi Arabia until August 1990. This choice of counterfactual abstracts from two effects whose omission could potentially bias the results. It is not straightforward to quantify either of these potential biases in the baseline counterfactual, but, since they are of opposite sign, they will at least partially offset one another.

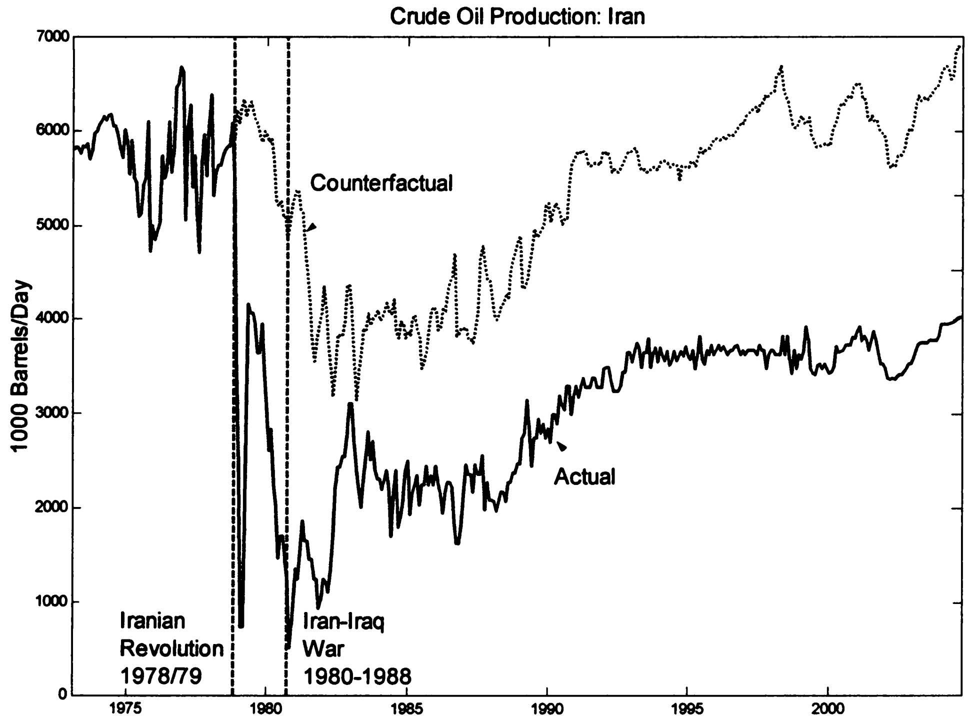
The first issue has to do with oil production responses to the outbreak of the Iran-Iraq War. IEA data show that, unlike in 1979, OPEC's capacity constraints were not binding in the early 1980s. Indeed, there was a temporary and moderate increase in oil production in the benchmark countries in the first months after the outbreak of this war, as shown in figure 2. In constructing the counterfactual for Iranian oil production for this period I ignore the possibility that the outbreak of the Iran-Iraq War through its effect on oil prices might have triggered this increase in oil production in the benchmark countries. To the extent that it did, this assumption would bias the results toward overstating the extent of the exogenous production shortfall.

Second, I make the important assumption that the global recession of the early 1980s that was responsible for large oil production cutbacks in the benchmark countries in 1980 and 1981 was driven mainly by a global monetary contraction rather than the delayed effects of the Iranian Revolution. Clearly this is a working hypothesis only and subject to further investigation. An alternative view would be that the Iranian Revolution of late 1978, through its effect on real oil prices, contributed to the economic downturn of the 1980s. Such a link has yet to be established. So far there is no direct evidence that the observed increase in the real price of oil in the second half of 1979 and in the first half of 1980 was causally related to the oil supply cuts during the Iranian Revolution. Not only the considerable delay in the oil price increases and their gradual ascent argues against this interpretation, but sustained

[7] Consistent with this widely held view, the *Oil and Gas Journal* (October 15, 1973, p. 44) summarized the purpose of the October 8 OPEC talks as follows: "The [oil] producing countries are seeking revision of the 1971 Tehran Agreement to take account of inflation and of increased product prices in world markets. The Tehran pact [which was initially supposed to be binding until 1976] provides for only 2.5% annual inflation." For a detailed discussion of this point, see Penrose (1976).

[8] To the extent that subsequent oil price increases were driven by the embargo and/or the war (as opposed to strong demand for oil), and to the extent that these price increases stimulated additional non-OPEC oil production over the period in question, the relative fall in Arab oil output after October 1973 may be inflated. We ignore this possibility given the existence of capacity constraints in most non-OPEC oil-producing countries that made short-term increases in oil output difficult. For example, the IEA reports that in 1973 world oil production exceeded the sustainable level of production by about 5%. For a more detailed discussion of these IEA data see section VI.

FIGURE 2.—IRAN



periods of oil price increases are consistent with a booming world economy not unlike that in 2004/05. Moreover, oil production data from the U.S. Department of Energy show that the Iranian Revolution was associated with only a small and temporary decline in world oil production. Finally, as shown in section VI, even conventional quantitative dummy measures of the exogenous oil supply shock fail to explain the bulk of the observed oil price increases between late 1979 and mid-1980. It is worth noting, however, that, to the extent that part of the reduction in oil production in the benchmark countries was due to the decline in real growth triggered by the Iranian Revolution, this would cause the shortfall implied by the baseline counterfactual to understate the true shortfall.

In August of 1990, the benchmark becomes total OPEC excluding Iran, Iraq, Saudi Arabia, and Kuwait, as Kuwait experiences an exogenous production shortfall of its own and must be dropped from the benchmark. Since much of the short-term production response to this event occurred in Saudi Arabia, as discussed in section IIIE, I again assume that production in the benchmark countries did not respond to the higher oil prices triggered by the Gulf War. While there is some increase in those countries' oil production in late 1990, that increase is also consistent with prewar production trends. In any case, starting in 1991, the path of oil production in these countries remains largely flat for several years, consistent with evidence of OPEC capacity limits starting in 1990.

Finally, following the civil unrest of December 2002, I also drop Venezuela from the benchmark. The resulting benchmark reflects the assumption that the observed increases in oil production in 2003 and 2004 were not stimulated by the Venezuelan crisis of late 2002 or the 2003 Iraq War. If this assumption were incorrect, this counterfactual would tend to overstate the exogenous shortfall of oil production in 2003 and 2004.

Figure 2 shows the complete counterfactual path together with the actual Iranian production levels. The vertical difference between these paths at each point in time may be viewed as a measure of the exogenous shortfall of Iranian oil production. Exogenous shortfall series of this type will be used below in constructing the measure of the exogenous OPEC oil supply shock. Before addressing this issue in section IIIG, I will discuss the counterfactuals for the remaining OPEC countries.

C. *Counterfactual for Iraq*

The counterfactual for Iraq begins in September 1978. Although Iraq was not directly involved in the Iranian Revolution, this exogenous event by releasing previous constraints set in motion a chain of events that culminated in the Iran-Iraq War of September 1980. Given the close relationship between the Shah and the United States and the military strength of Iran under the Shah, the idea of Iraq invading Iran prior to October 1978 would have been unthinkable. In fact, relations between the two countries were cordial to the point that Iraqi authorities imposed increasingly severe restrictions on the movements of Iranian dissidents in Iraq such as the fundamentalist Iranian Shia leader Ayatollah Khomeini, prompting Khomeini to leave Iraq for France in September of 1978 (see Terzian, 1985, p. 278). The fall of the Shah and Khomeini's seizure of power led to a break with the United States and a deterioration of Iranian military capabilities. In addition, the secular and socialist Baathist regime in Iraq felt threatened by the fundamentalist Shia movement in Iran, given its own majority Shia community. Thus, the Iranian Revolution provided both opportunity and added motive for Iraq to attack Iran.

The change in Iraqi policy toward its neighbor clearly would not have occurred, had the Iranian Revolution not happened. The Iraqi preparation for the coming war started immediately after the fall of the Shah. Iraq began stockpiling substantial stocks of arms and spare parts as well as foreign exchange reserves amounting to $35 billion right before the war. Just three weeks after the war broke out, Iraqi authorities boasted that they would be able to maintain the war effort for a year without exporting any oil at all (see Terzian, 1985, p. 282, for details).[9] The ability to build such reserves was crucial for Iraq since geography made Iraqi oil exports much more vulnerable than Iran's in case of a war. The only way Iraq could accumulate such reserves and pay for arms and spare parts was to export oil at an unprecedented rate. In this sense, the Iranian Revolution caused an increase in Iraqi oil production. It is this increased production in Iraq after September 1978 that motivated the exclusion of Iraq from the benchmark, when discussing the Iranian counterfactual. This reasoning further suggests that we view the increase in Iraqi oil production relative to other OPEC countries between October 1978 and August 1980 as an exogenously driven oil supply shock rather than a supply response to market conditions. Put differently, the preparations for the Iran-Iraq War were arguably an endogenous response to the Iranian Revolution, yet clearly exogenous with respect to global macroeconomic conditions.[10]

A suitable counterfactual for Iraq can be constructed by comparing actual Iraqi oil output to the production level that would have prevailed, had Iraqi production after September 1978 grown at the same rate as that of total OPEC minus the combined production of Iraq, Iran, and Saudi Arabia. The rationale for excluding Iran and Iraq from the benchmark is self-evident. I exclude Saudi Arabia given the evidence, discussed in section IIIE, that Saudi Arabia itself was subject to three temporary exogenous production shocks in 1978/79, 1990/91, and 2002/03. Moreover, as discussed in section IIIE, Saudi production decisions starting in mid-1979 were influenced by price developments within OPEC that in turn may have been driven in part by exogenous events. This observation suggests that it is safer to exclude Saudi oil production data from the baseline benchmark, although I will consider including Saudi production data as part of my sensitivity analysis. Finally, further adjustments to the counterfactual, consistent with the assumptions for Iran, were made for Kuwait starting in August 1990 and for Venezuela starting in December 2002, when those countries experienced their own exogenous production shortfalls.

Figure 3 plots the actual and counterfactual production levels. There is clear evidence of the disproportionate increase in Iraqi production after the Iranian Revolution. Accounting for this temporary increase in output is essential for assessing the shortfall following the outbreak of the Iran-Iraq War in September 1980. A substantial proportion of the actual fall in Iraqi output in 1980 is simply a reversal of the earlier positive shock, so conventional measures of oil shocks are bound to overestimate the exogenous fall in output. As in the case of Iran, oil output recovered in the early 1980s (as Iraq was able to make use of newly built pipelines beyond the reach of Iran) and exceeded the counterfactual even before the war had ended in 1988. As in the case of Iran, the enormous financial cost of the war is likely to have been the cause. Even after the war had ended, this buildup continued until the outbreak of the Gulf War, as Saddam Hussein aimed to recover his military strength. Figure 3 shows that Iraqi production remained at very low levels for several years following the defeat in the Gulf War, owing no doubt to the sanctions and export restrictions imposed by the United Nations. By 2000 Iraqi oil output had recovered to near-normal levels on average, as the effectiveness of sanctions had been eroded. The unprecedented volatility of Iraqi oil production after 1997 undoubtedly reflects the uneven enforcement of the U.N. sanctions over time. Iraqi oil output collapsed again in March 2003 with the outbreak of the Iraq War.

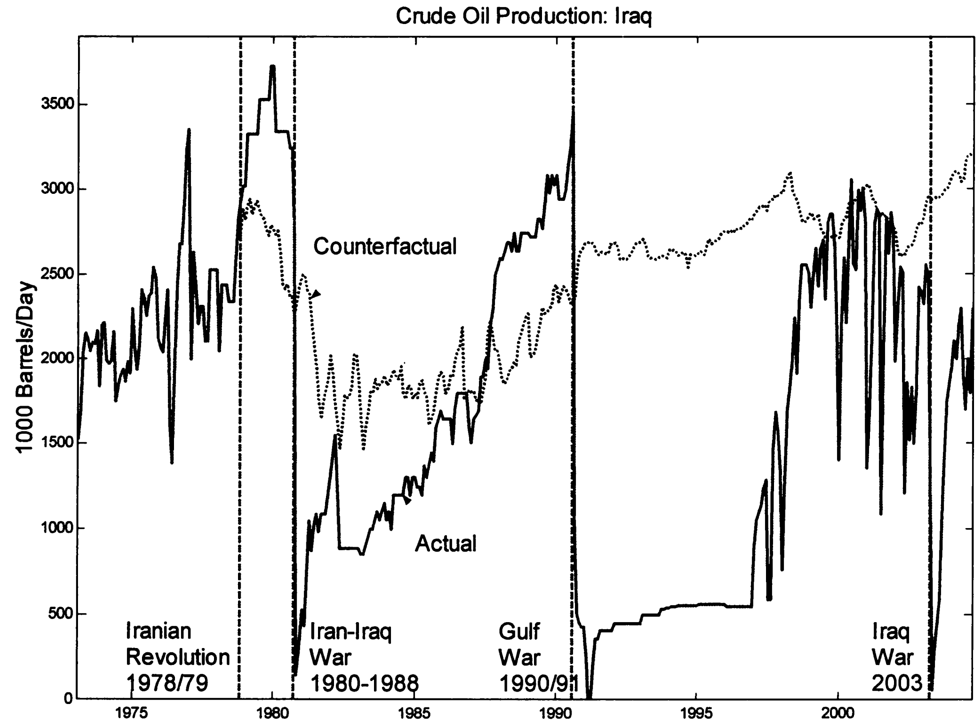
D. *Counterfactual for Kuwait*

After the 1973/74 episode, Kuwait was not affected directly by any exogenous event until the invasion of Kuwait in August 1990. A natural counterfactual for the remaining period can be based on the oil production of total OPEC minus the combined production of Kuwait, Iran, Iraq, and Saudi Arabia, with Venezuela added to this list after December 2002, when that country experienced its own exogenous production shortfall. The rationale for

[9] There are no publicly available data documenting this buildup of foreign exchange reserves. Since Iraq stopped reporting data on its foreign exchange reserves to the IMF in 1978, the series ends in December 1977.

[10] The shift in Iraqi policy in 1979 was not mirrored by similar preparations for war in Iran, which for the time being was preoccupied with the revolution. In fact, when Iraq launched a large-scale military assault in September 1980, the Iranian leadership was taken by surprise despite earlier warnings of Iraqi intentions. Part of the reason was that the Iranian army at this point had been rendered ineffective by the revolution; part that the Iranian regime was preoccupied with an internal power struggle, and part that the Iranian government considered news of the Iraqi war preparations a deliberate piece of Soviet misinformation, a reasonable response given the Soviet invasion of neighboring Afghanistan (see Terzian, 1985, pp. 278–281).

FIGURE 3.—IRAQ



this counterfactual follows from the earlier discussion. Figure 4 shows that by 1994 most of the production shortfall had been recovered, although production remained slightly below what one would have expected based on the benchmark.

E. *Counterfactual for Saudi Arabia*

Saudi Arabia differs from other OPEC oil producers in that on many occasions it had the ability to act as a swing producer. Over the period in question it also served on several occasions as a supplier of last resort. This poses special challenges for the construction of the counterfactual.

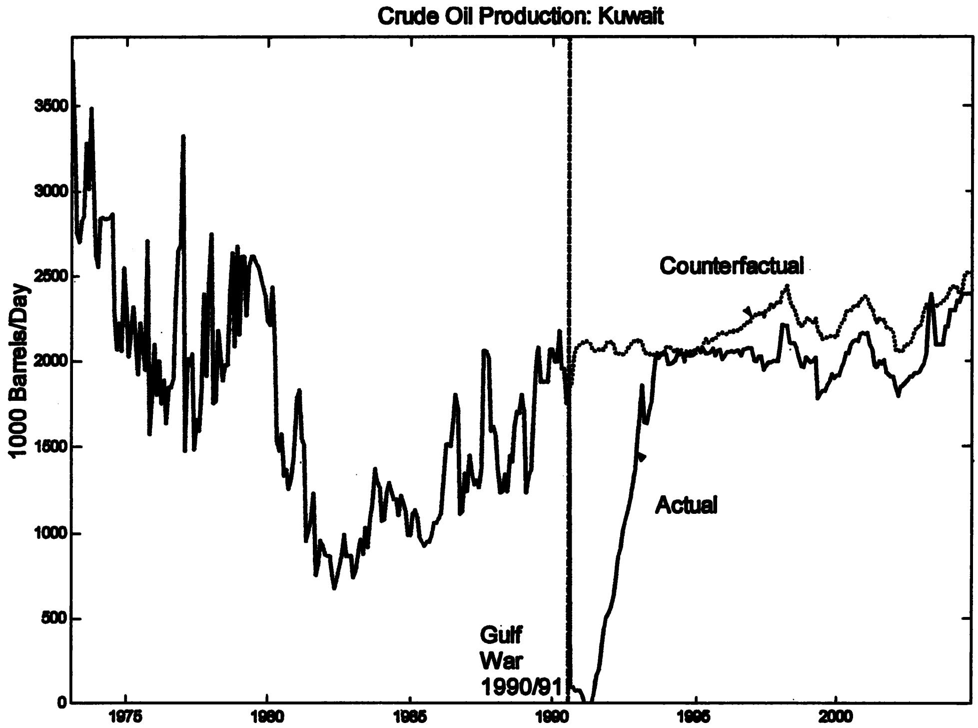
The unusual increase in Saudi oil production starting in late 1978 has already been mentioned. There were two distinct motivations for this increase. Initially, the Saudis increased production to offset the incipient shortfall caused by the Iranian Revolution. It was understood that this increase would be temporary and lapse as soon as the Iranian crisis was over. Indeed, the Saudis sharply curtailed their production in April of 1979.[11] Since this initial spike in Saudi oil production in 1978/79 was a discretionary act and a direct response to the Iranian Revolution (rather than to higher oil prices triggered by the Iranian Revolution), the case can be made that it must be treated as exogenous with respect to the U.S. economy. The Saudi counterfactual for the Iranian Revolution involves simply comparing its production during October 1978–April 1979 to that of other OPEC countries excluding Iran and Iraq. Since the response of Saudi production to the revolution—like the Arab oil embargo—is treated as strictly temporary, there is no need to model the counterfactual beyond April 1979.

In contrast, when oil prices accelerated again after April 1979, the Saudis increased production to force other OPEC members to bring oil prices down to levels deemed acceptable by the Saudis. This response was clearly not exogenous and cannot be treated as an exogenous disturbance of oil production. Another (smaller and less pronounced) increase in Saudi output coincided with the outbreak of the Iran-Iraq War, but there is no historical evidence that this increase was motivated by anything else than the power struggle within OPEC. For that reason I do not include this event either.

There are, however, two more exogenous events that prompted an increase in Saudi production. After the invasion of Kuwait in August 1990. Saudi Arabia temporarily increased its production with the aim of offsetting the shortfall created by the war. The obvious counterfactual consists of the growth of oil production in OPEC after

[11] Skeet (1988, p. 163) suggests that the April decision reflected the belief that the balance of supply and demand in the global oil market had reverted to an acceptable degree of normality given the Iranian production increase in March (also see Seymour, 1980, p. 183). Global oil production data lend support to the Saudi view. During the first two months of the Iranian Revolution global oil production actually increased. After the peak in November, there was a temporary fall from December 1978 until March 1979. From April 1979 on, global oil production levels exceeded the September 1978 level of 62.477 million barrels.

FIGURE 4.—KUWAIT



excluding Saudi Arabia, Iraq, Iran, and Kuwait from the total. Similarly, the 2003 Iraq War prompted a preemptive increase in Saudi production. The Iraq War differs from all earlier episodes in that the war was anticipated and without a well-defined end. Nevertheless, it is clear that Saudi Arabia temporarily increased its oil production to offset jitters in the oil market. A counterfactual may be constructed by focusing on the period from July 2002 when the war option was increasingly discussed in public until October 2003, when the focus had shifted from open war to the pacification of Iraq.

Figure 5 implies that all three episodes that have been identified based on historical evidence were associated with sharp positive spikes in Saudi oil output relative to the relevant OPEC benchmarks for each event, lending credence to the view that Saudi Arabia's response was caused by the exogenous events in question. It may be possible to identify smaller war-related peaks of a similar nature in the production data of other countries, but the magnitudes will be so much smaller that I abstract from this possibility.[12]

F. *Counterfactual for Venezuela*

The Venezuelan oil shock of December 2002 is not usually included in the list of exogenous oil shocks, but, as figure 6 shows, is a natural candidate for an exogenously induced oil shock. The civil unrest in Venezuela was followed by a sharp and well-defined reduction in crude oil output. The counterfactual is based on total OPEC production minus the combined production of Venezuela, Iraq, Iran, Kuwait, and Saudi Arabia, following the reasoning used in the analysis of the previous counterfactuals.

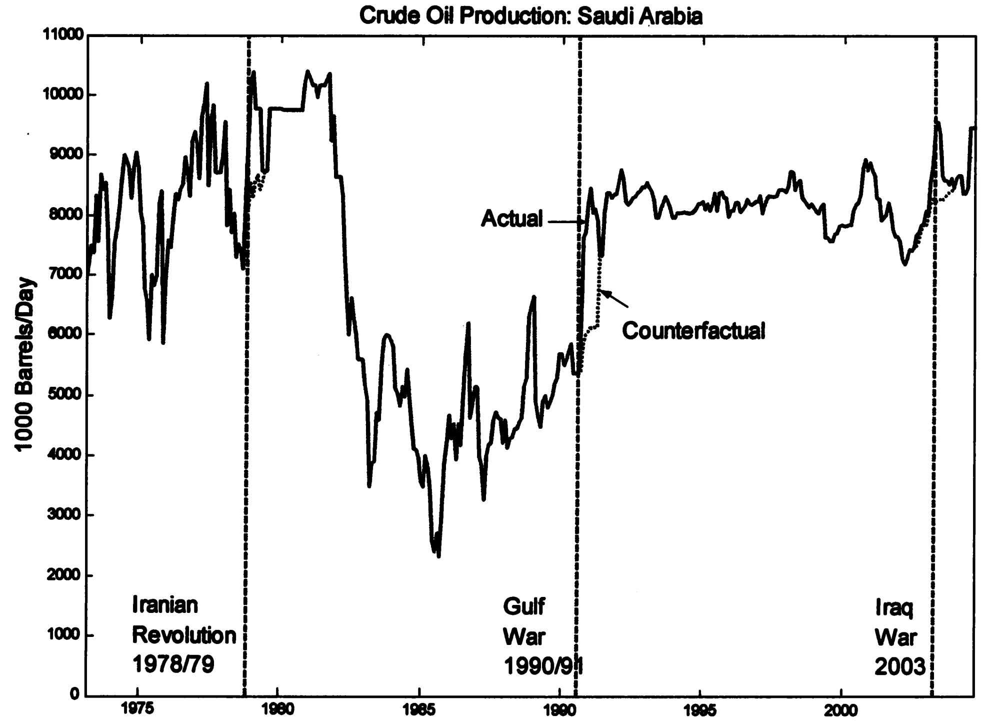
G. *Constructing a Historical Series of Exogenous OPEC Oil Supply Disturbances*

A measure of the OPEC-wide exogenous oil production shortfall may be obtained by summing the time series of country-specific shortfalls described in sections IIIB–IIIF and the OPEC-wide shortfall associated with the 1973/74 episode. I express this monthly production shortfall series as a share of world oil output. A natural measure of the exogenous OPEC oil supply shock is the change in the normalized production shortfall over time, shown in figure 7. For the remainder of the paper, I will treat this series as the baseline measure of exogenous oil supply shocks (aggregated to quarterly frequency where appropriate). It can be shown that the series in figure 7 is serially uncorrelated, consistent with the view that the shortfall series follows a random walk.

Figure 7 shows that exogenous oil production shocks range from almost +4% to almost −7% of world crude oil production at monthly frequency. Exogenous changes in crude oil production driven by political events in OPEC countries account for about 6% of the variability in world

[12] I also explored the alternative assumption that the Saudi production response might have been endogenous after all. The qualitative results in section IV are not affected by this question.

FIGURE 5.—SAUDI ARABIA



crude oil production changes, according to the baseline counterfactual. As expected, all oil dates with the exception of the 2002 Venezuelan crisis and the 2003 Iraq War are associated with large exogenous swings in oil production.

The movements in the baseline exogenous oil supply shock measure reflect key historical events. For example, the initially negative and then positive spikes exhibited by the baseline exogenous oil supply shock series in 1973/74,

FIGURE 6.—VENEZUELA

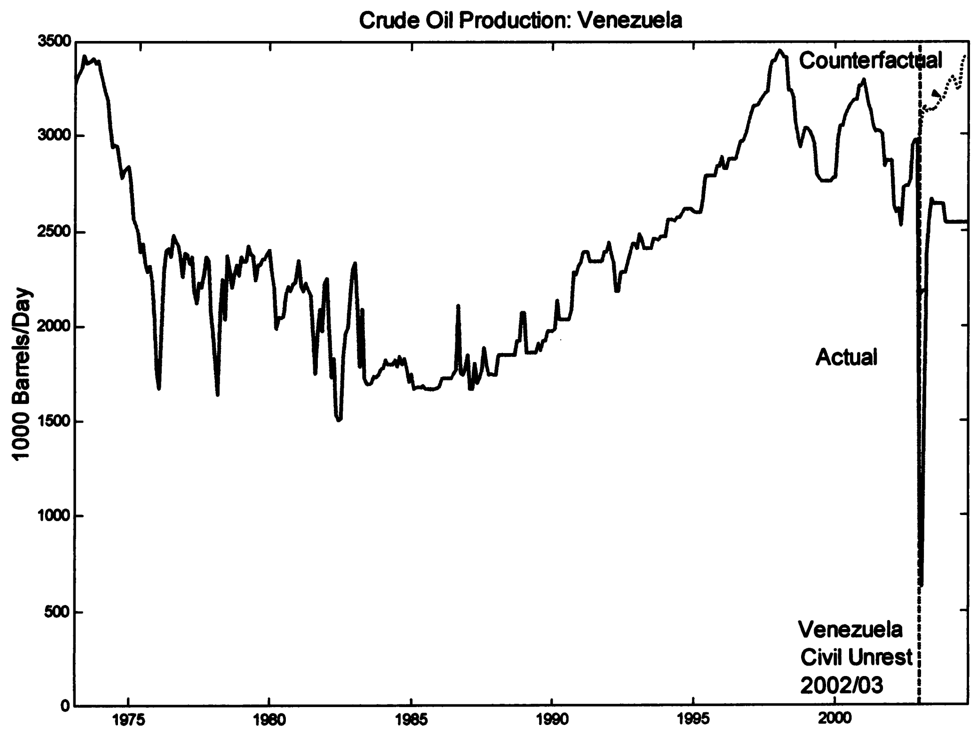
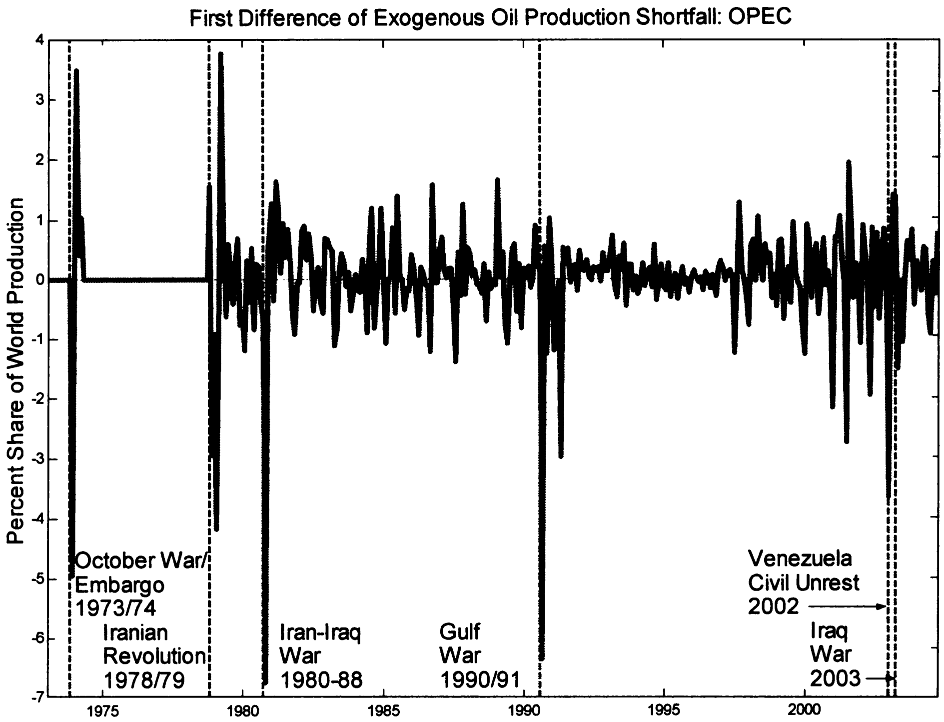


FIGURE 7.—EXOGENOUS OIL SUPPLY SHOCKS: OPEC



1978/79, and 1980/81 and to a lesser extent in 2002/03 make sense in that these periods are characterized by temporary exogenous oil production shortfalls. As the shortfalls are at least partially reversed, the initial negative change in production must by construction be followed by a positive change. Ignoring this reversal would amount to postulating a permanent reduction in oil supplies. The same is true for 1990–1995, except that both the decline and the reversal caused by the Persian Gulf War are more gradual, translating into a persistent decline, followed by a persistent increase in the shock measure.

In addition, there are two periods of high volatility in exogenous oil supply shocks: the 1980s and the late 1990s. The volatility after 1981 reflects mainly the fluctuations of Iranian and Iraqi oil production under wartime conditions. The shocks are predominantly positive over this period, because both Iran and Iraq were able on average to increase their oil output in an effort to finance the war, but with uneven success. The volatility in the late 1990s, in contrast, is largely driven by the fluctuations in Iraqi oil production, which in turn reflected the gradual erosion of the U.N. sanctions and their uneven enforcement. Between the aftermath of the Persian Gulf War and the resurgence of Iraqi oil production in 1997 the exogenous oil supply shock is effectively 0, as one would expect, given the absence of exogenous shocks during this interval.

The series in figure 7 forms the basis of the empirical analysis in the remainder of the paper. For the regression analysis in sections IV and V, this series is aggregated to quarterly frequency.[13] My main empirical findings based on this quarterly series are remarkably robust to alterations in the benchmark underlying figure 7. Specifically, I explored three more scenarios: (i) excluding the exogenous Saudi production response from the baseline; (ii) dropping the oil production shortfall associated with the 1973/74 episode from the baseline; and (iii) including Saudi Arabia in the benchmark starting in 1974. These changes did not affect the qualitative results reported in section IV below.[14]

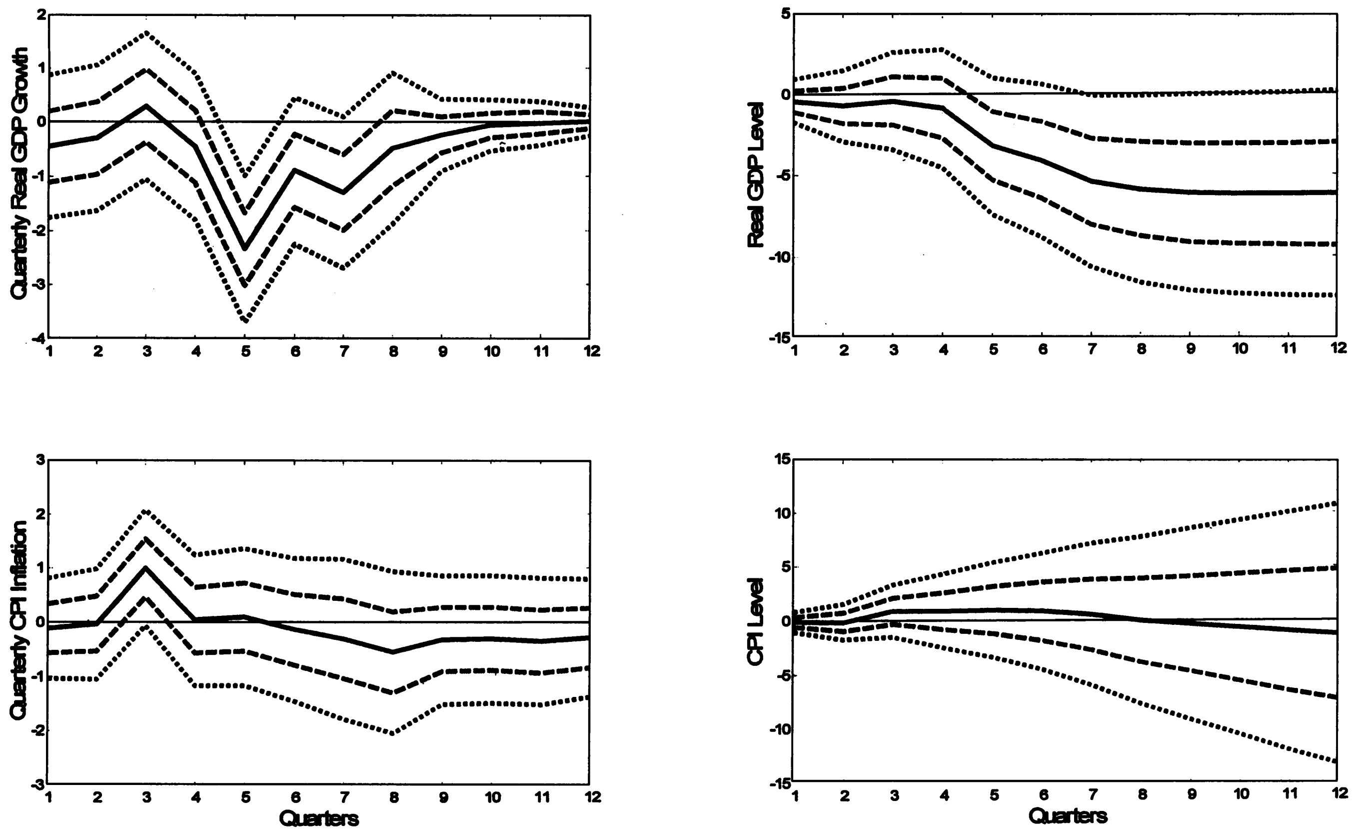
IV. The Dynamic Effects of Exogenous Variations in the Production of Oil on U.S. Macroeconomic Aggregates

An important question from a policy point of view is how exogenous oil production disruptions, as defined in section

[13] See section V for further discussion and plots of the quarterly series. The qualitative results are robust to using the monthly series when the use of either series is feasible.

[14] Yet another concern is that the counterfactuals may become less plausible, the further we extrapolate beyond the most recent exogenous event for that country. This concern is particularly relevant in the case of Iran. As part of my sensitivity analysis, I addressed this concern by returning Iran to the benchmark group in 1994.1, when Iranian oil production had stabilized following the recovery from the Iran-Iraq War. I also returned Kuwait to the benchmark in 1994.1, given that Kuwaiti oil production had returned to near prewar levels at that point, suggesting that the shortfall had been temporary (see figure 4). This amounts to setting the exogenous oil supply shocks in those countries to 0 starting in January of 1994. It also involves recomputing the counterfactual production levels for Iraq, Saudi Arabia, and Venezuela. The exogenous oil supply shock series arising from this fourth counterfactual looks very similar to the one shown in figure 7, once again supporting the robustness of my results.

FIGURE 8.—DYNAMIC EFFECTS OF A 10% WORLD OIL SUPPLY DISRUPTION: BASELINE COUNTERFACTUAL



III, affect CPI inflation and real GDP. Since GDP data are available only at quarterly frequency, for this section, I rely on quarterly analogues of the data shown in figure 7.[15] The analysis is based on an OLS regression of real GDP growth on a constant, lagged values of real GDP growth, and lagged values of the exogenous oil supply shocks.[16] Provided that the exogenous regressor in question is independent of all other potential exogenous regressors (a plausible assumption in the present context), we can think of the exogenous variations in oil supply as a "treatment," the effect of which can be measured by the dynamic response of real GDP growth to an exogenous shock in oil production. The latter response by construction will quantify the "tendency of the U.S. economy to perform poorly in the wake of . . . historical conflicts [in the Middle East]" (Hamilton, 2003, p. 364). Inference for this response is standard and OLS estimates are consistent. This type of regression approach provides the basis for the answer to the following two questions.

A. *How Does U.S. Real GDP Growth and CPI Inflation Respond to Exogenous Oil Supply Shocks?*

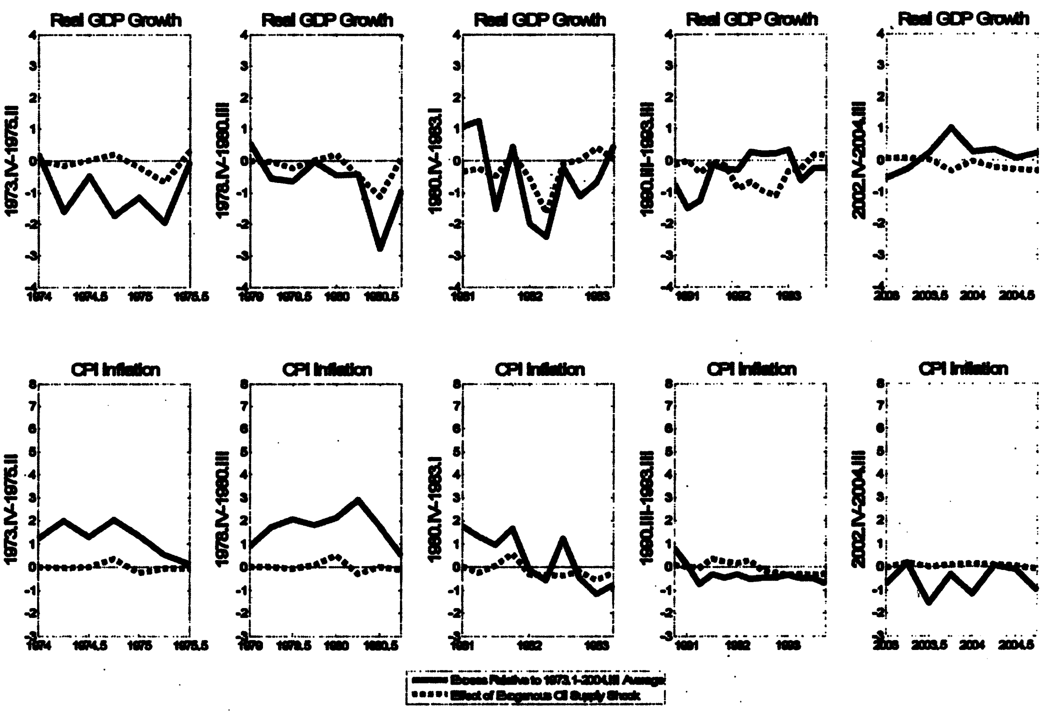
Point estimates of the dynamic effect of adverse exogenous oil supply shocks on real U.S. GDP growth and the level of real GDP are shown in the upper panel of figure 8. The estimates show a sharply negative growth rate five quarters after the oil supply shock, before the response reverts back to 0. This negative growth according to the 68% confidence band to some extent persists until the seventh quarter; according to the 95% confidence band it lasts only for one quarter. There is no discernible decline in the level of real GDP until the fifth quarter, and the decline in real GDP at longer horizons, while large, is very imprecisely estimated.

The same OLS regression approach may also be used to study the response of CPI inflation. Point estimates of the dynamic effects of exogenous oil supply shocks on U.S. CPI inflation and the CPI level (in the lower panel of figure 8) suggest a sharp spike in inflation three quarters after the shock. Otherwise the response is flat. The spike is significant at the 68% confidence level, but not the 95% confi-

[15] The quarterly shocks are computed as the change over time in the quarterly average of the normalized shortfall.

[16] This specification nests an autoregression in the dependent variable. In practice, I allow for four lags of the dependent variable, but eight lags of the exogenous oil supply shock measure. While four lags of the dependent variable will be sufficient to control for autoregressive dynamics, the extra lags for the oil supply shock are intended to allow for the possibility that oil supply shocks may take two years to affect the economy. Regressions that also included the contemporaneous value of the exogenous oil supply shock gave qualitatively similar results.

FIGURE 9.—COMPARISON OF REAL GROWTH AND CPI INFLATION EXPERIENCE BY OIL SHOCK EPISODE



dence level. At all other horizons the response of CPI inflation is insignificant at the 68% level. Moreover, the price level measured by the CPI does not increase significantly in response to the shock, even at the 68% level. The evidence that exogenous oil supply shocks over the post-1973 sample period have not caused sustained inflation is consistent with the theoretical arguments and related empirical evidence in Barsky and Kilian (2002, 2004).[17]

B. *How Would U.S. CPI Inflation and Real GDP Growth Have Evolved in the Absence of Exogenous Oil Supply Shocks?*

Impulse responses of the type shown in figure 8 effectively postulate a one-time permanent reduction in oil supplies. This thought experiment helps gauge the dynamic effects of exogenous shocks to oil production, but is at odds with the actual historical experience during events such as the Iranian revolution, the Iran-Iraq War, or the Gulf War in that these events involved production shortfalls that were at least partially reversed over time. In assessing the overall effect of these shortfalls it is essential that we consider the cumulative effects of the full sequence of exogenous oil supply shocks, some negative and some positive, as shown in figure 7. This may be accomplished by a historical decomposition of the data based on the regression model described earlier. The cumulative effect of exogenous oil supply shocks on real growth and inflation, respectively, may be computed by subtracting from their actual value the fitted value obtained by setting the exogenous oil supply shocks to 0 in the estimated OLS regression. These estimates naturally are subject to considerable sampling uncertainty, so the resulting counterfactuals are only suggestive.

Figure 9 addresses the question of how the U.S. economy would have evolved since the 1970s in the absence of exogenous oil supply shocks. For our purposes, it is useful to focus on five specific oil shock episodes, namely the 1973/74 Arab-Israeli conflict, the Iranian Revolution in late 1978, the outbreak of the Iran-Iraq War in late 1980, the Persian Gulf War of 1990/91, and the Venezuelan crisis and Iraq War of 2002/03. In assessing the relative contribution of exogenous oil supply shocks to macroeconomic performance, it is helpful to normalize the actual real GDP growth and CPI inflation outcomes relative to their long-run average such that a value of 0 corresponds to the "normal" level of real growth, and negative numbers indicate abnormally low real growth by historical standards. For each of the five major shock episodes, figure 9 plots these deviations of real growth and CPI inflation from their full-sample average as well as the estimated cumulative effect of the exogenous oil supply shocks.

[17] Responses from analogous regressions for core inflation (that is, consumer price inflation excluding food and energy) look qualitatively similar to the baseline results, except that the peak response in the third quarter is smaller than in figure 8 and that the response of core inflation is significantly negative in quarter 1 and quarter 8.

The upper panel of figure 9 suggests that the abnormally low growth of 1974/75 was not caused by exogenous production cutbacks in the Middle East. The effect of these cutbacks on real growth was negligible. The impact of exogenous oil supply shocks was somewhat larger in mid-1980 and especially in early 1982. This does not mean that exogenous oil shocks caused the recession of 1982. First, recessions are defined relative to some measure of the business cycle, whereas figure 9 plots real growth rates. Second, figure 9 does not imply that all of the abnormally low growth during that period can be attributed to the effects of oil. In fact, relative to its long-run mean of 0.75%, U.S. quarterly real GDP growth was abnormally low—for reasons not related to the exogenous oil production shortfall—in almost every quarter from early 1979 until late 1980 and again from mid-1981 until early 1983. For example, in mid-1980 quarterly growth would have fallen to −1.5% (relative to the long-run mean) even in the absence of exogenous oil shocks. There also would have been several quarters of negative real growth in 1981 and 1982 without these shocks. According to figure 9, exogenous oil supply shocks were at best a contributing factor at times. Specifically, exogenous oil supply shocks accounted for about two-thirds of the reduction in real growth relative to normal levels in one quarter of 1982 and for slightly more than one-half in one quarter in 1980. In the remaining quarters, the contribution of exogenous oil supply shocks to the abnormally low growth (to the extent that there was any contribution) was much smaller.

The upper panel of figure 9 also shows that exogenous oil supply shocks caused a substantial and persistent reduction in real growth in 1991/92, as one might have expected, but the bulk of that decline occurred only a full year after the invasion of Kuwait.[18] In contrast, following the 2002/03 shocks, there is hardly any evidence of a reduction in real growth being caused by exogenous oil supply cuts.[19]

[18] A possible explanation of the initial drop in real growth could be a temporary increase of fears of future oil supply disruptions in late 1990. Those fears never materialized and hence are not reflected in the production-based measure of the exogenous oil supply shock used in this paper. That explanation appears consistent with the behavior of the real price of oil, as discussed in section VI.

[19] Although the effect of exogenous oil supply shocks on real GDP growth is small, figure 9 shows that there is some effect. This raises the possibility that this reduction in real growth in turn lowered demand for oil and thus prompted oil producers to lower production. The maintained assumption in constructing the counterfactual has been to rule out such a feedback. Clearly, to the extent that the maintained assumption is wrong, the baseline counterfactual will understate the magnitude of the oil production shortfall in these periods. For the early 1980s, the output losses associated with exogenous oil supply shocks are mainly limited to one quarter each, following the Iranian Revolution and the outbreak of the Iran-Iraq War. Thus, the maintained assumption may be defended by observing that the countries in the benchmark group are unlikely to have responded much to such short-lived fluctuations in real growth. In contrast, in 1992 the estimated effect lasted for nearly a year. There is little evidence, however, of any reduction in oil production in the benchmark countries, and oil producers worldwide (according to IEA figures) continued to operate near their capacity limit.

To summarize, the evidence suggests that exogenous oil supply shocks made little difference overall for real GDP growth, although they did matter for specific episodes.[20] Even stronger results are obtained for CPI inflation. The lower panel of figure 9 shows that for each episode in question the increase in CPI inflation that can be attributed to exogenous oil supply cuts is negligible. In particular, there is no evidence that the abnormally high inflation rates of the 1970s and early 1980s can be attributed to the effects of exogenous oil supply cuts.

V. Quantifying Exogenous Oil Production Shortfalls: A Comparison with the Status Quo

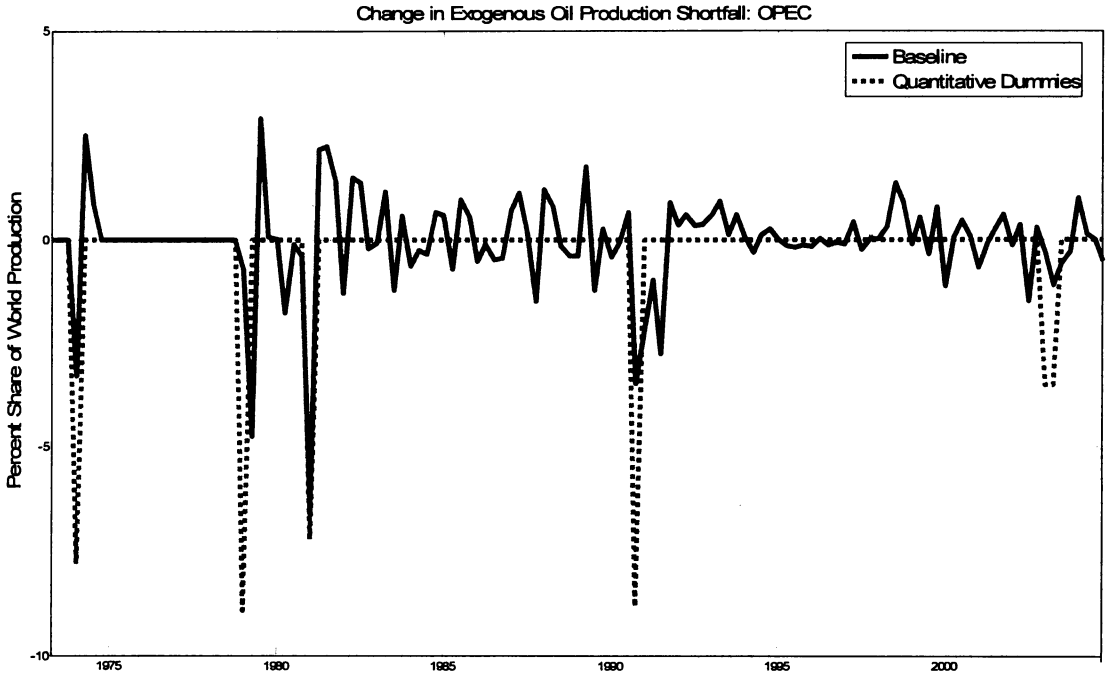
In this section, I will compare the measure of exogenous oil supply shocks proposed in this paper to an alternative approach that has been used in the literature and involves the use of quantitative dummies. This state-of-the-art approach to constructing series of exogenous oil supply disturbances is described in Hamilton (2003).

A. *Conceptual Comparison with the Quantitative Dummy Approach*

It is clear that *any* analysis of the shortfall will by construction involve a counterfactual. Hamilton's methodology is no exception. Thus, the virtue of the approach proposed in this paper is to make the assumptions that go into the counterfactual explicit and to foster a critical discourse about these assumptions. The differences and commonalities of the two approaches are as follows: Hamilton (2003) proposes to use the drop in a country's oil production following an exogenous event in this country as a measure of the magnitude of the exogenous shock to the supply of oil. The exogenous events in question are the same considered in this paper. As in this paper, Hamilton focuses on the oil-producing countries directly involved in the event in question, but he uses the level of oil production in the month prior to the exogenous event as the benchmark. He compares that level to the lowest level of production that is observed subsequently. The difference in physical production levels over the period in question is expressed as a share of the average world oil production in the year in which the exogenous event started (see Hamilton, 2003, p. 390) and attributed to the quarter in which the exogenous event occurred. This approach is in essence a quantitative version of the dummy variable approach used by Hoover and Perez (1994) to model oil supply shocks.

[20] The finding that exogenous oil supply shocks played some role in economic downturns, but do not account for a significant portion of U.S. real output fluctuations, is reminiscent of findings in the literature on the theoretical dynamic stochastic general equilibrium model. For example, Backus and Crucini (2000) concluded that oil supply shocks make only modest contributions to the business cycle (also see Kim & Loungani, 1992; Finn, 1995). It is important to keep in mind, however, that neither the definition of the oil supply shocks nor the definition of real output used in these studies is compatible with the definitions used in this paper.

FIGURE 10.—COMPARISON WITH HAMILTON'S QUANTITATIVE DUMMY SERIES



The method of quantifying exogenous production shortfalls proposed in this paper has six distinct advantages compared with the conventional approach based on quantitative dummy variables: (i) It does not impose the implicit assumption that the level of oil production would never have changed in the absence of the exogenous political event. (ii) It allows the response of oil production to the exogenous event to be immediate or delayed. (iii) It allows the response to be long-lasting. (iv) It allows the response to be time-varying. (v) It allows the response to an exogenous political event to be negative or positive, possibly changing sign over time.[21] (vi) The proposed method also avoids some ambiguities related to the timing of events and the magnitude of the shortfall that arise in the construction of quantitative dummy measures (see Kilian, 2005, for further discussion).

The methodological approach to constructing the counterfactual that I follow in this paper is not without its own shortcomings, however, because it as well involves some degree of judgment and assumptions that cannot always be verified, as illustrated by the discussion of the counterfactuals in section III. This fact makes it important to contrast the implications of the baseline counterfactual to those of alternative counterfactuals including the quantitative dummy approach.

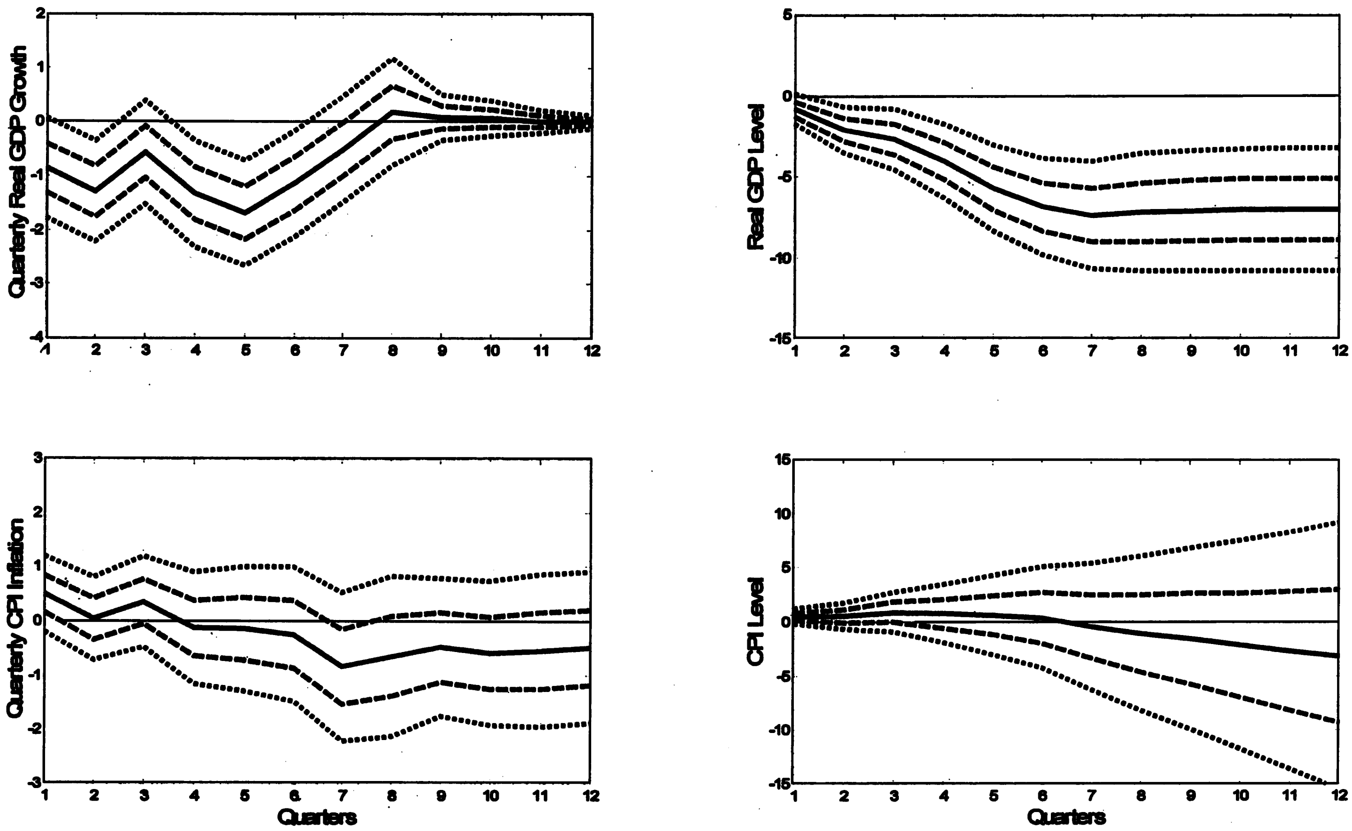
[21] This point is important as my analysis suggests that many exogenous oil production shortfalls were at least partially reversed over time, causing an initial negative shock to be followed by positive shocks. Moreover, the analysis in section III suggests that wars in the Middle East, after an initial decline in production, may exogenously stimulate higher oil production, when the parties involved resort to oil exports to finance the war.

B. *Empirical Comparison of the Shock Measures*

Since the quantitative dummy measure of exogenous oil supply shocks is constructed at quarterly frequency, it must be compared with the quarterly analogue of the monthly measure of the baseline exogenous production shortfall series shown in figure 7. Compared with the monthly shortfall measure in figure 7, this quarterly exogenous oil supply shock series is much smoother, as one would expect. Figure 10 plots the exogenous oil supply shock together with a suitably updated measure of Hamilton's (2003) quantitative dummy measure. The procedure used in constructing the Hamilton series follows exactly the description in Hamilton (2003). It makes sense to compare the quantitative dummy measure to the exogenous oil supply shock series (rather than the production shortfall) because Hamilton's (2003) intention is to capture the "initial shock" in the country where the event takes place (p. 391), while discounting all other exogenous variation in oil output triggered by that event.

Figure 10 illustrates important qualitative and quantitative differences between these measures. For example, the quantitative dummy series always stays below 0, even during times when production recovered for exogenous reasons such as in the course of the Iran-Iraq War or toward the end of the Arab oil embargo of 1973/74. Moreover, the quantitative dummy series indicates shocks of potentially very different magnitudes and persistence. The quantitative dummy measure typically records a larger shock following exogenous events than the baseline measure; in some cases more than twice as large as the baseline measure. The only

FIGURE 11.—DYNAMIC EFFECTS OF A 10% WORLD OIL SUPPLY DISRUPTION: QUANTITATIVE DUMMIES



exogenous event for which both measures indicate a shock of similar magnitude is the outbreak of the Iran-Iraq War in 1980.

C. *Direct Estimates of the Dynamic Effects of the Quantitative Dummy Measure of Exogenous Oil Supply Shocks on U.S. Macroeconomic Aggregates*

Figure 11 highlights the differences in the responses estimated using the two measures of exogenous oil supply shocks, while controlling for the method of estimation. It shows impulse responses that are obtained from least squares projections on lags of the quantitative dummy measure of exogenous oil supply shocks. The results are directly comparable to figure 8. The first difference is that the quantitative dummy measure suggests an earlier and more sustained reduction in real GDP growth than the baseline measure. It is also worth noting that the maximum response in figure 11 is only about -1.5% compared to -2.2% in figure 8. In addition, the quantitative dummy measure suggests an immediate and sustained decline in the level of real GDP that is significant at the 95% level, in sharp contrast to the delayed and insignificant decline in figure 8.

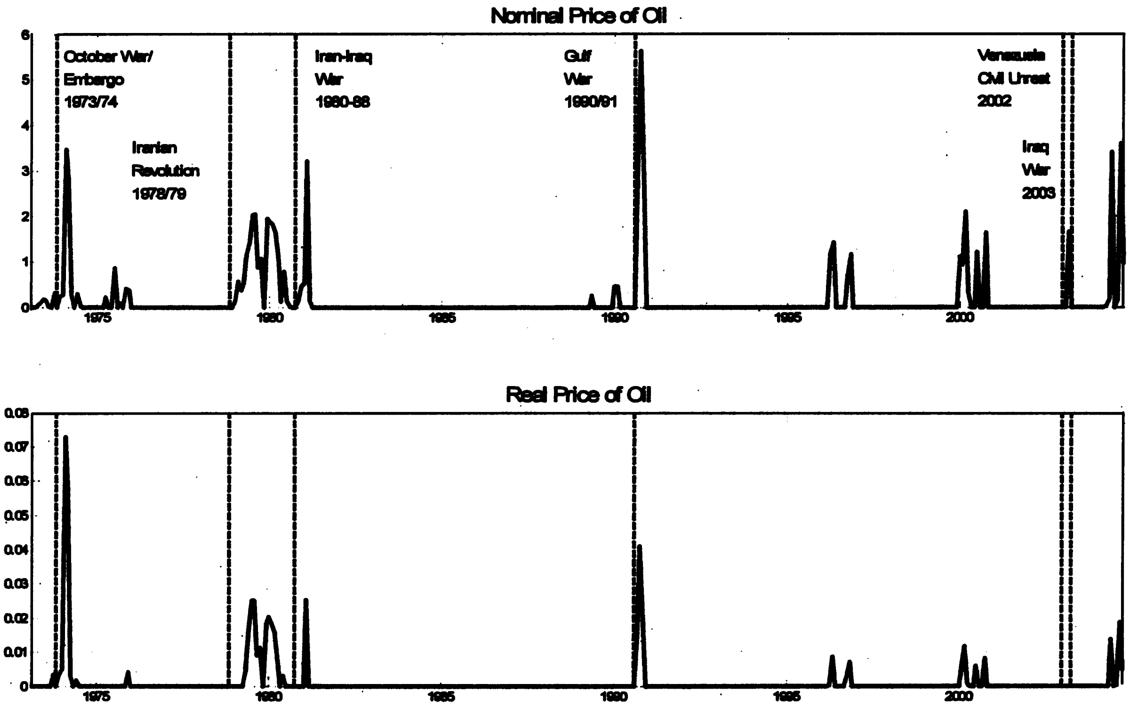
Unlike the baseline results, the response of CPI inflation in the lower panel of figure 11 shows no evidence of a spike in CPI inflation at any horizon. None of the responses are significant even at the 68% level except for a negative response after seven quarters. While the point estimate is positive for the first three quarters, the maximum increase is only about 0.4% (compared with a maximum of almost 1.0% in figure 8). The corresponding response of the level of the CPI is more negative at higher horizons than in figure 8, but equally insignificant at all horizons. These results illustrate that the definition of the exogenous oil supply shock measure greatly affects the qualitative and quantitative features of the estimated responses.[22]

VI. The Tenuous Link from Exogenous Oil Supply Shocks to Oil Price Shocks

There is an important distinction between exogenous oil supply shocks on the one hand and oil price shocks on the other. Early work on the effect of exogenous oil supply

[22] Whereas the estimated responses of overall CPI inflation are very sensitive to the definition of the exogenous oil supply shock, the results for core CPI inflation are more robust. The only differences are that (i) the negative impact effect is smaller than in the baseline model and statistically insignificant at the 68% level, (ii) the peak in quarter 3 is smaller than in the baseline model, yet remains significant, and (iii) the significantly negative core inflation rate found in the baseline model after quarter 8 already occurs in quarter 7.

FIGURE 12.—OIL PRICE SHOCKS MEASURED BY NET OIL PRICE INCREASE (RELATIVE TO PREVIOUS THREE YEARS)



shocks on U.S. real GDP was facilitated by the unique institutional arrangements in the oil market in the postwar period prior to 1973. Hamilton (1983) made the case that nominal oil prices over this period were effectively exogenous. None of these arguments apply to the oil market since 1973, however, and there is widespread recognition that today oil prices must be considered endogenous with respect to global macroeconomic conditions. Recently the case has been made that, nevertheless, nonlinear transformation of the price of oil designed to capture "oil price shocks" effectively identify the exogenous component of the price of oil. From our point of view these measures are of interest, because they seem to provide a simple alternative to the approach proposed in this paper. As the analysis below will demonstrate this is not the case.

A. *Nominal Oil Price Shocks*

Oil price shocks refer to unusually high oil prices relative to recent experience. Building on work by Mork (1989), Lee, Ni, and Ratti (1995), and Hamilton (1996), Hamilton (2003) proposed a formal definition of oil price shocks based on the net increase in the nominal price of oil relative to the maximum of the price of oil over the previous three years. This definition has been used in studying the responses of U.S. sectoral and macroeconomic aggregates to oil price shocks (see, for example, Bernanke, Gertler, & Watson, 1997; Davis & Haltiwanger, 2001; Lee & Ni, 2002; Hamilton, 2004).

Figure 12 applies this definition to the refiner acquisition cost of imported crude oil as defined by the Department of Energy (and extrapolated backwards until January 1972 based on the wholesale price index of crude oil). The upper panel of figure 12 shows that by this definition major nominal oil price shocks occurred in 1973/74, 1979/80, 1981, 1990/91, 2000/01, 2002/03, and 2004. Additional minor nominal oil price shocks occurred in 1975/76, 1989/90, and 1996/97. Hamilton (2003) suggested that the predictive power of these net oil price increases for macroeconomic aggregates such as U.S. real GDP growth can be attributed to their ability to filter out influences on oil prices that do not come from exogenous political or military events abroad.[23] This interpretation is contradicted by the fact that not every oil price shock in figure 12 can be associated with an exogenous political event in the Middle East (indicated by vertical lines). This is true not just for the relatively minor shocks, but also for some of the more important spikes such as those in 2000/01 and in 2004. Given that oil price shocks are not necessarily preceded by exogenous Middle Eastern political events, it is not clear that the observed increase in oil prices after those exogenous events was actually caused by those events.

[23] For example, Hamilton (2003, pp. 391, 395) writes that nonlinear transformations of this type "filter out many of the endogenous factors that have historically contributed to changes in oil prices" and "seem in practice to be doing something rather similar to isolating the exogenous component of oil price changes."

B. *Real Oil Price Shocks*

My analysis has focused on nominal oil prices so far, following the convention in the literature. From an economic point of view it is, of course, the real price of oil that matters for resource allocation. The distinction makes a difference in defining the oil price shock. The lower panel of figure 12 is based on the same oil price data as the upper panel, but deflated by the U.S. CPI. The first observation is that nominal oil price shocks are not necessarily real oil price shocks. Second, there is no real oil price shock at the time of the Venezuelan crisis in late 2002 or of the 2003 Iraq War, demonstrating that exogenous oil supply shocks need not be followed by real oil price shocks. Third, as in the nominal case, there are oil price shocks that are not immediately preceded by exogenous events, notably in 1996/97, 1999/2000, and 2004. Fourth, there are dramatic changes in the relative importance of oil price shocks. For example, the 1974 event is much more important than the 1991 event in real terms, whereas focusing on nominal data causes a reversal of this relationship. Fifth, the timing of oil price shocks may be affected by the distinction between nominal and real prices. For example, the oil price shock after the Iranian Revolution in real terms occurs only in 1979, not in late 1978 as the nominal data would suggest. I will focus on the real price of oil in the remainder of the paper.

C. *Evidence from Other Industrial Commodity Prices*

As discussed by Barsky and Kilian (2002, 2004), sharp increases in the real price of oil may occur even in the absence of exogenous events, for example, when increases in the demand for oil occur in the presence of capacity constraints in the oil market, which might explain the existence of oil price shocks at dates not related to exogenous events in the Middle East. By the same token one cannot simply attribute to exogenous events in the Middle East oil price shocks that occurred after those events because at least part of the observed oil price shock may have been driven by increased demand for oil and would have occurred even in the absence of that event. This point may be illustrated by applying Hamilton's net price increase method to the linearly detrended Commodity Research Bureau series of real non-oil industrial commodity prices.[24] Specifically, that application demonstrates that the 1973/74 and 1978/79 oil dates were associated with substantial net increases in non-oil industrial commodity prices, although these commodity price shocks had nothing to do with exogenous events in the Middle East, but were driven by global demand pressures that affected all industrial commodities including oil (see National Commission on Supplies and Shortages, 1976).[25] While we are far from fully understanding fluctuations in industrial commodity prices, this example illustrates by analogy the danger of spuriously associating exogenous events with net increases in oil prices.

D. *The Role of Capacity Constraints in Crude Oil Production in Amplifying the Effect of Demand Shifts on the Price of Oil*

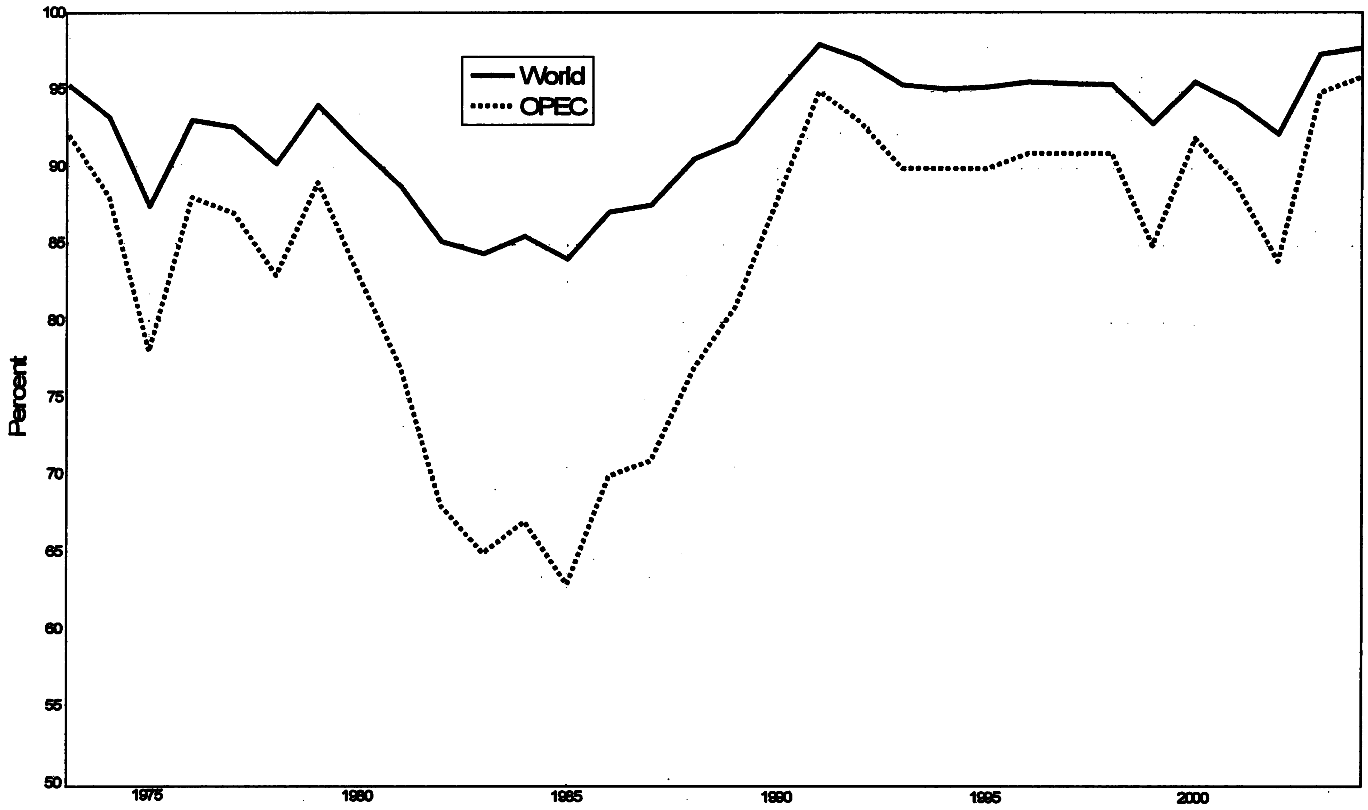
The magnitude of the price shock in industrial commodities in the early 1970s and late 1970s suggests that the bulk of the 1973/74 and 2004/05 oil price shocks (and at least a substantial component of the 1979/80 oil price shock) may have been due to rising demand for oil rather than political events in the Middle East, whereas for the oil dates of 1980 and 1990/91 there is no evidence of aggregate demand pressures in industrial commodity markets. Higher demand for oil is primarily driven by fluctuations in global economic growth. While these demand shifts are not large enough by themselves to explain large increases in the price of oil, combining such demand shifts with a near-vertical oil supply curve can generate a large increase in the price of oil. An essential ingredient in this explanation is the presence of capacity constraints in crude oil production in 1973, 1979, and 2004/05.

The IEA compiles annual data on OPEC capacity utilization since 1973. These data are not in the public domain. Moreover, a breakdown by country does not exist. Systematic data collection started only in 1990, while the earlier data are at best educated guesses. Nevertheless, these data are informative. Figure 13 shows IEA estimates of capacity utilization rates for OPEC as well as world oil production. The two series are compiled from different sources and their level is not necessarily compatible. In interpreting these data, it is useful to keep in mind that there is always a gap between the nominal capacity of an oil production facility and maximal sustainable capacity, with the latter usually running near 90% of the former. In practice, one never knows exactly how much production the system can handle on a sustained basis until a trial has been conducted (see Seymour, 1980, p. 169). The danger is that exceeding the threshold of sustainable production may permanently damage an oil field. With this in mind, we can think of capacity utilization rates near 90% as effectively operating at full capacity. Figure 13 shows that there indeed were capacity constraints both in OPEC and worldwide in 1973, 1979, and 2004/05, but no constraints in 1980/81, consistent with the discussion in the oil literature and the assumptions made in

[24] The data were downloaded from the Commodity Research Bureau (CRB) at http://www.crbtrader.com/crbindex. The industrial commodities include burlap, copper scrap, cotton, hides, lead scrap, print cloth, rosin, rubber, steel scrap, tallow, tin, wool tops, and zinc. Qualitatively similar results are obtained based on the CRB spot price index for all non-oil commodities. The detrending is required because of the long-term downward trend in non-oil commodity prices.

[25] The figure is not shown to conserve space, but is available in Kilian (2005).

FIGURE 13.—ESTIMATES OF CAPACITY UTILIZATION RATES IN CRUDE OIL PRODUCTION BY YEAR



the counterfactuals of section III.[26] There also is evidence that OPEC and world spare capacity had been steadily shrinking in the period leading up to the Persian Gulf War of 1990/91, and became binding with the onset of the war. Likewise in 1976/77 OPEC was nearing its capacity limit.

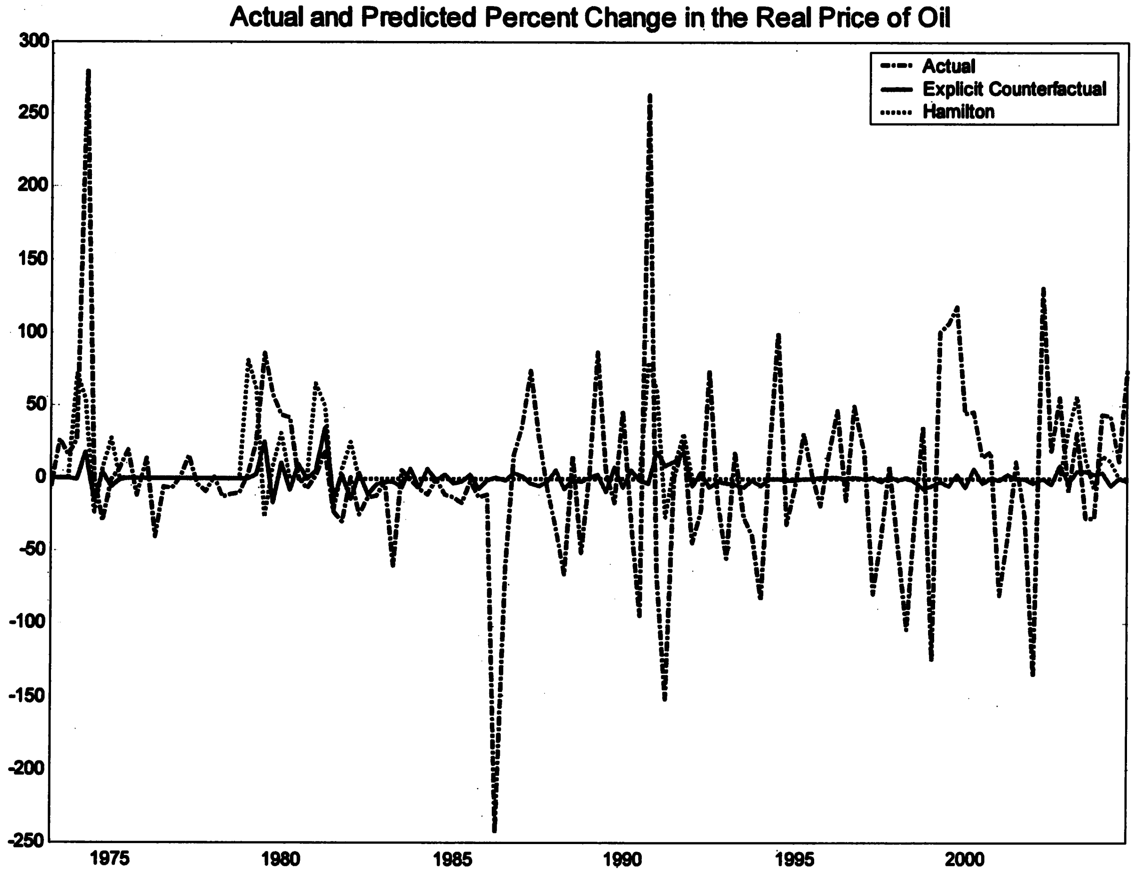
It is important to keep in mind that only the conjunction of capacity constraints and increases in the demand for oil will generate large oil price increases. For example, OPEC and world capacity utilization rates were very high in 1976/77, but demand for oil remained moderate and oil prices did not respond. In contrast, the increase in oil prices in 1973, 1979, and 2004/05 was preceded by an unusual buildup of global demand. It is important to keep in mind that the price of oil is determined in international markets and depends on global macroeconomic conditions, rather than just U.S. macroeconomic conditions. The buildups of demand for oil in 1973 and 1979 (and the drop in demand in the early 1980s) were noteworthy because at these times (but not at other times) the U.S., European, and Japanese business cycles were almost perfectly synchronized.

The 1973 demand shift was larger than subsequent demand shifts because it was further amplified by institutional changes in the oil market. Not only did real GDP for the developed world in the four years from 1970 until 1973 grow at 3.2%, 3.6%, 5.5%, and 6.3%, respectively (see Darmstadter & Landsberg, 1976, p. 17), but this shift in the demand for oil was compounded by the fact that prior to October 1973 the price of crude oil was regulated based on the Tehran-Tripoli agreements. This not only stimulated demand, for a given unit of real GDP, but, when the price ceiling came off, the price jumped to account for the cumulative demand pressures that had built up, not just the current demand pressures. Combining such a shift with a near-vertical oil supply curve was bound to generate a large increase in oil prices.[27]

[26] Since the 1973 observation refers to an annual average of the peak loads in the middle of the year, and the reduced load at the beginning of and toward the end of this year, we may infer that the capacity utilization rate in September of 1973 must have been much higher than 92%.

[27] Of course, not all of the oil price increase in 1973/74 is necessarily due to the demand increase. There presumably also were some increases in precautionary demand for oil (see, for example, MacAvoy, 1982, p. 46). Nevertheless, we have some benchmark for how much of an increase in oil prices one would have expected. For 1973/74 we know that other industrial commodities (for which there was no supply shock or increase in precautionary demand) showed similar increases as the price of oil (namely a tripling). Examples include scrap metal, paper and pulp, and lumber. The National Commission on Supplies and Shortages of 1976 concluded that these price increases were not driven by supply shocks. The only difference is that for other commodities prices were not regulated, so the price increases started earlier than for crude oil. If strong demand can have such effects on other industrial commodities, it is plausible that it also would be capable of moving the price of oil this much (see Barsky & Kilian, 2002, 2004). The plausibility of such magnitudes also is supported by the extent of the oil price increases in 2003–2005, which by all accounts were demand driven.

FIGURE 14.—ACTUAL OIL PRICE CHANGES AND OIL PRICE CHANGES EXPLAINED BY EXOGENOUS OIL SUPPLY SHOCKS



Similarly, the most recent years in the sample have been characterized by robust growth not just in the United States, but worldwide. This effect was compounded by the emergence of new oil importers in Asia that increasingly relied on oil rather than other forms of energy. In contrast, there is no evidence of unusually strong global real economic growth, when oil production reached capacity in 1990/91.

These examples illustrate that demand-driven oil price shocks only occur under very specific circumstances. For example, a strong expansion in the United States alone may not be enough to generate a sharp increase in oil prices, when this expansion is cushioned at the global level by a recession in other parts of the world. Moreover, any such buildup in global demand for oil must be seen in relation to existing capacity in oil-producing countries, and capacity constraints are not fixed over time. Whether or not the price of oil increases sharply depends on whether the shift in demand is large enough to hit the capacity constraint. This point is widely discussed in the oil literature (see Mabro, 1998). In subsequent work, I have proposed methods for measuring global demand for industrial commodities directly. The results are supportive of the explanation outlined here (see Kilian, 2006).

E. How Well Do Measures of Exogenous Oil Supply Shocks Explain Real Oil Price Increases during Crisis Periods?

Having shown that oil price shocks as measured by the net oil price increase are not necessarily related to exogenous oil supply shocks, I now turn to the related issue of measuring the extent to which percentage changes in the real price of oil can be attributed to current and lagged values of the exogenous oil supply shock. For this purpose I project the quarterly percentage change in the real price of oil on an intercept, the current value, and four lags of the exogenous oil supply shock. For a similar approach applied to the change in nominal oil prices see Hamilton (2003).

Actual and fitted real oil price changes. Figure 14 plots the fitted values for the two alternative oil supply shock measures discussed in section V against the actual percentage change in real oil prices. It may be tempting to conclude that the measure with the better fit in figure 14 would be the better measure of the exogenous oil production shocks, but it should be clear that this conclusion would not be correct. If, for example, a given shock measure mistakenly attributed the effects of rising demand for oil because of a global economic expansion to an exogenous event (such as a war), that measure would have higher R^2 by construction, but it would be the wrong measure to use. Thus, we cannot infer from the fit of the models in figure 14 which shock measure is more plausible; we only can discuss the implications of each measure for the real price of oil.

First consider the quantitative dummy measure of exogenous oil supply shocks. This measure predicts only about 20% of the change in oil prices that occurred in 1973/74, suggesting that the remainder is not related to exogenous oil production shocks. The baseline measure proposed in this paper assigns an even less important role to the production shortfalls associated with the Arab oil embargo. Either way we conclude that exogenous changes in oil production do not seem to have played a major role in the observed price increases during that episode. This finding is consistent with the earlier evidence that oil prices during this episode behaved similarly to other industrial commodity prices that were not subject to exogenous supply shocks.

For 1979/80, the quantitative dummy series not only suggests that the oil price increase should have occurred well before it actually did, but it also predicts a sharp spike, whereas the actual oil price increase in 1979/80 was more gradual. It is hard to imagine how this effect could have been offset by other factors, causing the actual data to show no spike at all at that point. The baseline measure, in contrast, indicates a much smaller effect of exogenous oil supply shocks, the timing of which is consistent with the timing of the actual oil price increase. For 1980/81 once again the quantitative dummy measure predicts a peak earlier than the actual data. Both exogenous oil supply shock measures overpredict the magnitude of the 1980/81 oil price increase, but the quantitative dummy measure much more so than the measure proposed in this paper. The fact that this oil supply shock had less of an effect than predicted may be attributed to the global economic slowdown at the time.

For 1990/91, the quantitative dummy measure provides a better fit with the sharp increase in actual oil prices, but even that measure predicts only about one-third of the actual oil price increase in 1990/91. In contrast, the baseline measure suggests that production shortfalls played a minor role in explaining the oil price increases of 1990/91. Moreover, according to the baseline measure, the production shortfalls affected the price only with a delay and caused sustained oil price increases rather than a spike. Either way, a substantial part of the 1990/91 oil price shock was not driven by the exogenous production shortfall. Finally, for 2002/03, neither measure of the exogenous oil supply shock does a good job at predicting the observed pattern of oil price changes. The quantitative dummy measure predicts higher oil price increases than occurred in 2002/03.

Apart from illustrating the timing problems encountered with the quantitative dummy approach, the regression evidence in figure 14 underscores that neither measure of exogenous oil production disturbances does a good job at explaining the magnitudes of the major oil price increases since 1973. For example, even the quantitative dummy measure explains only one-fifth of the 1973/74 spike and less than a third of the 1990/91 spike in oil prices. What then could possibly explain the additional sharp increase in the price of oil during these and other episodes? I have already outlined the important role played by shifts in the global demand for oil driven by higher global growth in conjunction with capacity constraints. There is strong evidence of demand for oil hitting capacity constraints in 1973, as evidenced by the data in figure 13 as well as the rapid and sustained increase in tanker freight rates long before the exogenous event in question (also see Seymour, 1980, p. 100). There is similar evidence for 2004/05, when strong demand due to the business cycle coincided with additional demand for oil from Asian countries that had not traditionally competed for this resource, and to a somewhat weaker extent for 1979. On the other hand, there is no evidence that the global business cycle in conjunction with capacity constraints played a role in driving up the price of oil in 1980/81 and 1990/91.

An alternative and complementary explanation is that concerns about the future availability of oil supplies, especially in the face of high demand for oil, could trigger a sudden and potentially large increase in precautionary demand for oil and hence in the price of oil, to the extent that supply is inelastic. There is casual evidence of such panic buying both in late 1973 and in 1979, for example. The latter phenomenon reflects fears of yet unrealized oil supply disruptions rather than concerns about already realized exogenous production shortfalls. While typically linked to exogenous events, shifts in precautionary demand may be highly idiosyncratic and differ from one episode to the next. They will be captured by the regression evidence underlying figure 14 only to the extent that they are linearly related to the actual exogenous oil supply disruptions that took place since 1973.

Beyond exogenous production shortfalls: The role of uncertainty about future oil supplies. There is evidence that fears about future oil supplies that are not linearly related to actual oil production cuts can play an important role at times. The 1990/91 episode in particular is instructive because for that period there is no reason to believe that global economic growth increased demand for oil enough to generate a sharp increase in oil prices; yet exogenous oil production shortfalls do not explain this episode well.[28] This

[28] The poor predictive performance of the exogenous oil supply shock measures during 1990/91 cannot be explained by estimation error. If we were to select coefficients that ensure that this price spike can be explained by the exogenous oil supply shock alone, then by construction we would end up with an even worse fit during most other exogenous supply shock episodes. Thus, there must be other factors at play that are not related to oil production shortfalls.

is true for the baseline measure of exogenous oil supply shocks as well as the quantitative dummy measure. Even the latter measure explains less than a third of the observed oil price increase in 1990. The only plausible alternative explanation is increased fears of future supply disruptions, not simply in the countries directly affected by the Persian Gulf War, but in other countries such as Saudi Arabia as well. The discrepancy between the smaller and more sustained increase in oil prices predicted by the baseline measure, in particular, and the observed sharp spike suggests the existence of a large uncertainty premium.

If a positive spike in oil prices is driven by increased uncertainty about future oil supplies, it should be followed by a negative spike when the uncertainty is resolved. Indeed there is evidence in figure 14 of such a reversal after the end of the 1991 Persian Gulf War. This evidence suggests that a shift in uncertainty rather than the physical supply disruption caused by these wars may explain the bulk of the observed sharp and immediate spikes in oil prices that are left unexplained by exogenous oil production disturbances.[29] The uncertainty-based explanation seems plausible in that the 1990/91 episode was fundamentally different from any other episode in our data in that the invasion of Kuwait for the first (and only) time in history created an imminent military threat to the oil supplies of Saudi Arabia. The United States responded to this threat by moving troops to Saudi Arabia, but only in November of 1990 had enough troops been assembled to neutralize the threat to Saudi oil fields (see United States Central Command, 1991). This unprecedented shift in uncertainty may be captured by adding a dummy variable to the regressions underlying figure 14 that takes on a value of 1 for the third quarter of 1990, when the imminent threat to Saudi Arabia first arose, and of -1 for the fourth quarter, when it was eliminated.

It is important to note that the quantitative dummy approach is not designed to capture the effects of these shifts in uncertainty on the price of oil. Figure 14 demonstrates that, as an empirical matter, the quantitative dummy approach does a poor job at predicting the change in the real price of oil in 1990/91. This finding is not surprising as there is no reason for the observed cutback of Iraqi and Kuwaiti oil production in 1990/91 to have an effect on the price of oil that is proportionate to that of increased fears about future Saudi oil supplies. Moreover, the quantitative dummy approach is not designed to capture the sharp decrease in the price of oil that is an essential feature of the uncertainty-based explanation of changes in the price of oil in 1990. Finally, if such a shift in expectations is specific to the 1990/91 episode, as I have argued, excluding the dummies will bias the regression estimates. Thus, including dummies designed to capture the shifts in uncertainty is essential, regardless of the measure of exogenous production cutbacks used.

After adding these dummies, the R^2 of the two regressions becomes very similar, suggesting that the regression based on quantitative dummies in figure 14 mistakenly attributes some of the effects of uncertainty about future Saudi oil supplies to exogenous production cutbacks in Iraq and Kuwait. Including the current value and one lag of this dummy variable in the regressions greatly improves the fit of *both* models in question for the 1990/91 episode.[30] Upon including the dummies, there is little to choose between the two measures of exogenous oil supply shocks for that period, suggesting that the earlier regressions confounded the effect of exogenous oil production shortfalls in Iraq and Kuwait in 1990/91 with the effect of the imminent threat to Saudi oil supplies. Similarly, for the 1980 oil supply shock, the two measures now give virtually identical results. The timing problem of the quantitative dummy measure in this episode vanishes. Both exogenous supply shock measures overpredict the effect of the oil supply shock on the price of oil in 1981, consistent with the view that a global economic slowdown largely offset the adverse effects of the 1980 oil supply shock. For the 1979/80 episode, the baseline exogenous oil supply shock measure now predicts a much larger increase in the price of oil, the timing of which is consistent with the actual data. The baseline shock series accounts for three-quarters of the increase in the price of oil in the third quarter of 1979, but not for any of the subsequent increases. The quantitative dummy series implies an even larger increase in the price of oil, but the timing of the predicted increase is premature. Similarly, for the 1973/74 episode both exogenous supply shock series now predict a much larger oil price increase. Nevertheless, they only account for a third and a sixth, respectively, of the actual oil price increase in 1973/74, reaffirming the earlier result that much of the observed oil price increase for this episode cannot be attributed to exogenous oil supply shocks. For the 2002/03 episode there is no improvement in fit. Exogenous oil supply shocks do little to explain the observed oil price changes.

To sum up, the analysis so far suggests three conclusions: (i) In general, exogenous oil production shortfalls are of limited importance in explaining oil price changes during crisis periods. This result is robust to the choice of the exogenous oil supply shock measure. (ii) The predictions of the baseline exogenous supply shock measure are easier to

[29] For the earlier episodes the evidence on the role of uncertainty is less clear-cut. Exogenous events such as major wars or revolutions in the Middle East should have a fairly immediate effect on the price of oil, if they cause a major increase in uncertainty about future supplies of oil. This was not the case after the Iranian Revolution in October 1978 or the outbreak of the Iran-Iraq War in September 1980, for example. It took more than half a year before prices started taking off after the Iranian Revolution and the process continued into 1980. Moreover, to the extent that these events increased uncertainty about future oil supplies, that uncertainty dissolved gradually, as these events dragged on, making it difficult to separate the uncertainty effect from other factors.

[30] The figure is not shown to conserve space, but is available in Kilian (2005).

reconcile with the actual oil price data than the predictions of the quantitative dummy measure in that their timing in 1979 is more consistent with the actual data. (iii) Of the episodes studied, only the 1980/81 oil price increases can be attributed to exogenous oil supply disruptions, but these increases were small by historical standards. Exogenous oil supply disruptions were an important contributing factor for the oil price increases in the third quarter of 1979, but do not help explain the subsequent oil price increases during that episode. Exogenous oil supply disruptions explain only a small fraction of the oil price increases during the 1973/74, 1990/91, and 2002/03 episodes.[31]

VII. Conclusion

Since the oil crises of the 1970s there has been strong interest in the question of how oil production shortfalls caused by wars and other exogenous political events in OPEC countries affect oil prices, U.S. real GDP growth, and U.S. CPI inflation. This paper focused on the modern OPEC period of the oil market since 1973. I proposed a new methodology for quantifying exogenous oil supply shocks and contrasted it with the conventional quantitative dummy approach. I also implemented a new approach to measuring the dynamic effects of exogenous oil supply shocks on macroeconomic aggregates and on the price of oil.

The aim has been to make explicit the assumptions underlying the counterfactual. There is no doubt that the baseline assumptions outlined in this paper could be improved and refined. It is worth stressing that there is nothing in the methodology outlined here that would a priori systematically bias the results in favor of assigning a smaller (or for that matter a larger) role to exogenous oil shocks or to their effects on oil prices and GDP growth. In fact, the estimated peak responses of U.S. macroeconomic aggregates to exogenous oil supply shocks in some cases are larger than estimates obtained using the conventional quantitative dummy approach. Nevertheless, large impulse responses do not necessarily translate into a large contribution of exogenous oil supply shocks to economic fluctuations since historically negative exogenous oil supply shocks tend to be offset at least in part by subsequent positive shocks.

It may seem surprising that political events in the Middle East do not have a larger impact on macroeconomic aggregates. One reason is that the approach in this paper has been causal, albeit reduced-form, and hence incorporates the average effects of endogenous responses of oil production elsewhere in the world and of U.S. policymakers. Another part of the answer is that this paper has followed the existing literature on the effects of exogenous oil supply shocks in focusing on the role of physical supply interruptions. My analysis suggests that exogenous oil production shortfalls narrowly defined can account for only a comparatively small part of oil price movements. This does not necessarily mean that exogenous political events in the Middle East are of lesser importance, however. Even if physical production does not move, expectations of future oil supply interruptions alone may have powerful effects, as is evident in my analysis of the 1990/91 Persian Gulf War episode. Such expectations are not directly observable, but a natural extension of the current line of work would be to identify observables other than oil production data that are likely to drive fears of future oil supply disruptions. Finally, my analysis provided indirect empirical support for the notion that the oil price increases in 1973, 1979, and 2004/05 were driven by strong global demand for oil in conjunction with capacity constraints in crude oil production. The important task of quantifying fluctuations in global demand for oil driven by real growth (and of assessing the relative importance over time of each of the competing explanations proposed here) is taken up in a separate paper (see Kilian, 2006).

[31] I also conducted this analysis using monthly data with twelve lags of the exogenous oil supply shock series and two lags of the dummy variable, dated as August 1990 and December 1990. Using monthly data allows one to capture more precisely the dynamics of the price adjustment. The results were remarkably similar to the quarterly results. The main difference is that the counterfactual provides a somewhat better fit for the 1980/81 episode.

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